Power to Attend: School Electrification and Student Attendance in Rural Uruguay

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October 2026

Abstract

Does grid connection raise attendance at rural primary schools? In 1,102 rural public primary schools in Uruguay, 1992 to 2019, a difference-in-differences design compares schools when they connect with schools in the same region not yet connected. A grid connection raised the share of students attending 141 days or more by about 4 percentage points over the first two years with electricity (standard error 1.23), and lowered the share attending 71 to 140 days by 3.9 points. The gain's sign is more robust than its size. Most of the gain comes from schools connected in 1993 to 2002, before Plan Ceibal's laptops reached rural schools. Solar panels show no detectable gain, but estimates are too imprecise to rule out effects as large as the grid's. Enrollment, teachers, and closure do not change detectably. At census resolution, hamlets next to a school gained electricity about when it connected; nighttime lights register hamlets and villages, not schools. To our knowledge, this is the first Latin American estimate, against a comparison group, of how a school's own electrification affects attendance.

Keywords: school electrification; rural education; student attendance; difference-in-differences; Uruguay.

JEL codes: I21, I25, O18.

1 Introduction

In 2023, more than one in five primary schools in the world had no electricity, and in the least developed countries the share was more than half (United Nations Statistics Division, 2026). Governments and international programs close this gap by extending the grid to schools or, where the grid does not reach, by installing solar panels. Kenya used both in its public primary schools (Koima, 2024), and Lights to Learn, a program of the Organization of Ibero-American States (OEI), installed solar panels in 556 rural schools of thirteen Latin American countries between 2011 and 2018 (Organización de Estados Iberoamericanos para la Educación, la Ciencia y la Cultura, 2018).[1] Both approaches rest on the premise that a school with electricity is a better place to learn and a more attractive place to attend. Yet the case for that premise is made mostly with cross-country correlations and anecdotes (United Nations Department of

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[1] Argentina, Bolivia, Colombia, Costa Rica, El Salvador, Guatemala, Honduras, Nicaragua, Panama, Paraguay, Peru, the Dominican Republic, and Uruguay.

Economic and Social Affairs, 2014), and a review of the field acknowledges how little evidence stands behind it (Welland, 2017).

So far, the evidence that electricity improves children's schooling is sobering. Randomized subsidies that connected rural Kenyan households to the grid had no detectable effect on children's test scores (Lee et al., 2020). Making Indian villages eligible for a national electrification program, or extending the grid to localities in Burkina Faso, did not detectably change primary enrollment (Burlig and Preonas, 2024; Schmidt and Moradi, 2026). And the evaluations that take the school itself as the unit find small effects or none (Seo, 2017; Berkouwer et al., 2018; Koima, 2024). A natural reading is that electricity, by itself, does little for education.

This reading may be premature, because the existing evidence looks at margins and settings in which the effect of a school's electricity is hard to see. Examination results and enrollment plausibly depend on inputs accumulated over years, whereas attendance can respond to the conditions students find at school every day; yet none of the school-level evaluations above measures it (Seo, 2017; Berkouwer et al., 2018; Koima, 2024). Where the grid reaches a whole locality, much else changes with it. In Burkina Faso, the grid brought new drinking-water systems and brighter nights along with electricity for some of the local schools, so the estimates capture the arrival of electricity in a place rather than in a school (Schmidt and Moradi, 2026). We therefore ask a narrower question: when a rural school gets electricity, do more of its students attend?

Uruguay lets us answer this question with records that cover the rural public primary schools of an entire country over three decades. Rural schools were small and isolated. The median school had 15 students, and three in five had a single teacher. Nearly four in five had no census locality, hamlet or village, within a kilometer, and the median school lay 23 kilometers from a locality of 5,000 inhabitants or more. In 1992, about half of these schools had no electricity. Over the following decades, the state-owned electric utility, the National Administration of Power Plants and Electrical Transmissions (UTE), connected the remaining schools to the grid at different times, and schools the grid had not yet reached received solar panels from the same utility; in 2013–2014, Lights to Learn replaced those panels in the 78 schools still off the grid. Of the 1,039 schools open in 2021 whose source of electricity we know, all but 15 were on the grid, and those 15 had a Lights to Learn panel.

We use this rollout to ask whether a grid connection raises the share of students who reach satisfactory attendance,[2] and whether it changes repetition, enrollment, or teachers. We also ask whether solar panels, the technology used where the grid did not reach, deliver similar gains; when any gains arose and whether they depended on the laptops that Plan Ceibal delivered to every public primary school student between 2007 and 2009 (Yanguas, 2020); and what changed around a school once it was connected.

The literature answers these questions only in part. Evidence on electrifying schools is scarce, and evidence that isolates the school is scarcer. In a review of studies published between 1990 and 2010, the positive association between school electricity and student outcomes disappears in the higher-quality studies (Glewwe et al., 2013). Only one of the 66 evaluations of rural electrification in a 2020 systematic search used the school as the unit of analysis (United Nations Economic and Social Commission for Asia and the Pacific, 2021). In a randomized trial in 164 Tanzanian secondary schools, the estimated first-year effect of solar panels with television sets on pass rates in the national examination was 2.8 percentage points over a control mean of 58.2 percent, with a 95 percent confidence interval from -1.1 to 6.7 points (Seo, 2017). In Kenya, connecting secondary schools to the grid did not detectably change the number of students who

[2]Satisfactory attendance is 141 days or more in the school year, about 76 percent of the school days a student should attend (ANEP–CEP, 2008). The Uruguayan Administration of Public Education (ANEP) counts attendance of 71 to 140 days as insufficient and of 70 days or fewer as intermittent dropout (ANEP, 2010).

sat the exit examination or their scores (Berkouwer et al., 2018). In Kenyan public primary schools, solar installations were followed by about 1 percent more students sitting the national examination and grid connections were not, and neither detectably changed scores or enrollment (Koima, 2024). In Latin America, studies compare Brazilian schools with and without electricity, exploit municipal eligibility rules in Brazil, or follow the district rollout of rural electrification in Peru; they record enrollment, dropout, or whether children attend school, but not how often they attend, and none isolates the school from its locality (Mejdalani et al., 2018; López Cajiao, 2018; Dasso et al., 2015).

A larger literature studies household electricity. Comparisons of connected and unconnected households associate it with more schooling and more study time (Khandker et al., 2009a,b, 2014; Aguirre, 2017; Alam and Kaneko, 2019), but households choose whether to connect, and designs that address that choice find little. Beyond the null results above, solar lanterns lent to students in rural Bangladesh raised attendance at surprise school visits by 8.4 percentage points over a control mean of 54.4 percent (Kudo et al., 2019). The effect, however, weakened over the year, was no longer significant at the third visit, eleven months after the lanterns arrived, and did not carry over to promotion or grades. Across six studies, a systematic review of off-grid technologies finds no statistically significant effect on attendance (Yavuz et al., 2025).

Electricity need not reach schools and households at the same time. How quickly households connect once electricity reaches an area varies, and the price of a connection matters (Lee et al., 2020). In the communes of a World Bank project in Vietnam, most of which got the grid between 2002 and 2005, the share of connected households rose from 8 to 81 percent in three years; according to the authors, communes helped poor households pay the connection fee (Khandker et al., 2009b). In Kenya in 2014, 89 percent of public facilities but only 32 percent of households had electricity (Lee et al., 2020). In Burkina Faso, where the grid electrified some of the local schools in the year it arrived, household access reached about 20 percent six years later (Schmidt and Moradi, 2026). Telling the effect of a school's electricity apart from that of electricity in its students' households requires observing both.

The literature also leaves open whether the technology matters. Solar panels can reach schools that the grid does not, but the evidence comparing them with grid connections comes mostly from households that adopted one or the other in different places. In Ethiopia, the schooling of children in households with solar catches up with that of children in households on the grid within two years, and in Senegal the effects of the two cannot be told apart (Lavecchia et al., 2025; Magbondé et al., 2025). To our knowledge, the only study that estimates the effects of the two on schools separately covers three years in Kenya (Koima, 2024). Whether electricity matters through the devices it powers is just as unsettled. In Uruguay, the quasi-experimental evidence on Plan Ceibal's laptops finds mixed effects on test scores, no gain in the years of schooling completed by early adulthood, and no estimate for attendance (Ferrando et al., 2011; Yanguas, 2020).

This paper makes three contributions. First, we provide evidence on how electrifying schools affects how often students attend, a margin that none of the school-level evaluations reviewed above measures. With records that cover the rural public primary schools of an entire country, we follow each school from 1992 to 2021 (attendance until 2019, before the pandemic) and observe the shares of its students with satisfactory attendance, with insufficient attendance, and in intermittent dropout, so we can see where in the attendance distribution the change occurs. To our knowledge, no previous study in Latin America estimates, against a comparison group, how electrifying the school itself affects how often its students attend. Studies for Brazil and Peru instead estimate how electrifying the locality affects whether children attend school (López Cajiao, 2018; Dasso et al., 2015).

Second, we show that the attendance gains from electricity did not wait for laptops. Pro-

grams such as Lights to Learn deliver electricity together with computers and connectivity (Organización de Estados Iberoamericanos para la Educación, la Ciencia y la Cultura, 2018). Lights to Learn's own evaluation could not build a comparison group and relies on teachers' perceptions (Organización de Estados Iberoamericanos para la Educación, la Ciencia y la Cultura, 2018); to our knowledge, ours are the first estimates of its effects, for the schools where it replaced an earlier solar panel, although they are too imprecise to interpret. A review of rural school electrification devotes one of its four chapters to the information and communication technologies that electricity makes possible (Welland, 2017), and a study of rural electrification in Brazil attributes its gains to electricity and technology inside the school (López Cajiao, 2018). Yet the gains we estimate come mostly from schools connected between 1993 and 2002 (285 of the 337 schools that changed), years before laptops reached rural classrooms; for connections after 2002, the estimate is too imprecise to tell whether the effect was smaller.

Third, we trace what happened around a school once it was connected, which school-level evaluations rarely observe. Linking school records to the 1996, 2011, and 2023 population censuses and to nighttime satellite imagery, we follow electricity from the school to the households and hamlets around it. Our estimates therefore capture the effect of a school's connection together with the changes that followed around it. Most schools, however, stood apart from any settlement, although the grid could still reach scattered households around them. We test whether the gain reflects children trading work for school, and we show that nighttime lights, which studies of electrification use as an outcome (Schmidt and Moradi, 2026; Figueiredo Walter and Moneke, 2024), register hamlets and villages rather than schools.

Our analysis rests on a school-year panel of Uruguay's rural public primary schools from 1992 to 2021. It links ANEP's annual school records (attendance in the three bands, repetition, enrollment, and teachers) with the date on which each school was connected to the grid or received a solar panel, compiled from the electric utility's and ANEP's records, and with census localities, gridded population, and satellite imagery that describe each school's surroundings.

The design exploits the staggered timing of electrification. The grid did not arrive at random. It reached first the schools closer to the existing network and to lit areas, in more populated places, and with more students (Appendix Table 5). We therefore compare each school that changed its source of electricity with schools that started in the same state and had not yet changed, within the same region and year. We use the estimator of de Chaisemartin and D'Haultfœuille (2020, 2026), which remains valid when effects differ across cohorts and over time. Placebo estimates for the years before electrification test whether the two groups followed parallel paths. We wrote down the main specification before estimating it.

Connecting a school to the grid raised the share of its students reaching satisfactory attendance by about 4 percentage points over the first two school years with electricity (3.95, standard error 1.23). On average, 84 percent of students reached it in school-years without electricity. The gain survives a pairs bootstrap, a permutation of connection dates, and trends that vary with schools' initial characteristics, and it stays positive, though less precisely estimated, in later years. Its size is less robust than its sign. Dropping the enrollment weights or the 1994 cohort (the one that weighs most), or keeping only the connection dates of high certainty, gives 2.4 to 3.0 points, and not all of these estimates are significant at the 5 percent level.

This gain mirrors a fall of 3.9 points in the share with insufficient attendance, while the share in intermittent dropout, about 2 percent of students, did not detectably change. Electricity thus moved irregular attenders up to satisfactory attendance rather than bringing back the few in intermittent dropout. Enrollment, the number of teachers, and school closure did not detectably change, nor did connections detectably draw students from nearby schools not yet on the grid, although with 29 close pairs we cannot rule out a moderate shift. Repetition shows no effect we

can interpret, because its trends before the connection differ between connected and comparison schools. Solar panels produced no detectable gain, but their estimates are too imprecise to rule out effects as large as the grid's; moving from no electricity to any source raised the share reaching satisfactory attendance by about 3.5 points. Sixth graders in schools connected in 1992–1996 scored 0.04 to 0.27 standard deviations higher in the 1996 national assessment than those in schools still without electricity, but with 43 schools none of these differences is significant at the 5 percent level.

Electricity also reached households around the schools. Between the 1996 and 2011 censuses, the share of households with grid electricity rose 31 percentage points more in census zones whose first school was connected to the grid between the censuses than in zones whose schools were still off the grid in 2011, nearly all of them with a solar panel. In hamlets next to the school, dwellings had almost no grid electricity in the censuses taken before the school's connection and nearly all had it in those taken after, so that, at the resolution of the censuses, they were connected at about the same time as the school. The censuses, though, observe few schools just before or just after their connection, and monthly night lights show that the timing varies. In the census zones whose school was connected between the censuses, the share of young adults aged 18 to 24 who had completed lower secondary school also rose 10 to 11 points more, a suggestive result.

The attendance gain is present in isolated schools as well, with no hamlet or village within a kilometer (3.2 points, standard error 1.3); in schools next to one, the estimate is larger (6.1, standard error 2.6), but too imprecise to tell the two apart. Among isolated schools, the estimate for solar panels (1.4 points, standard error 3.9) cannot be told apart from that for the grid. The gain is larger, and lasts longer, far from localities of 5,000 inhabitants or more (6.6 points, standard error 1.9, against 1.1 closer), although the difference does not survive the pairs bootstrap or a correction for the splits we tested. The attendance gain was not detectably larger where more children worked in 1996: the bootstrap confidence interval rules out that it was more than 3.2 points larger there, although not a smaller difference.

Section 2 describes rural schools in Uruguay, their electrification, and the data (Section 2.3). Section 3 presents the empirical strategy, and Section 4 the results. Section 5 examines the channels through which electricity could have raised attendance, and Section 6 concludes.

2 Context and Data

2.1 Rural schools in Uruguay

Uruguay had about 3.5 million inhabitants in 2023 (INE, 2026); the World Bank classified it as an upper-middle-income country until 2013 and as a high-income country since then (World Bank, 2024), and it has some of the lowest levels of poverty and inequality in Latin America (World Bank, n.d.). Since 2007, education has been compulsory from age 4: two compulsory years of initial education precede primary school, which covers ages 6 to 11 (Uruguay, 2007; ANEP, 2025). Between 2000 and 2013, 83 to 88 percent of students in grades 1 to 6 were enrolled in public schools (ANEP, 2014).

We follow the national classification for rural schools by ANEP (ANEP, 2025). Rural schools are many but small. By the beginning of the 1990s, ANEP classified 1,246 schools as rural; the average one had about 22 students in grades 1 to 6, and three in five had a single teacher.[3] As the

[3]Own calculations from ANEP's school records, for the schools ANEP classified as rural in each year. Among all the schools ANEP classified as rural, the median one had 14 students in 1992, 9 in 2010, and 7 in 2021; in our estimation sample, the median school had about 15 students in 1992–1994. Appendix Table 1 describes the

rural population declined, so did enrollment: the average rural school had 15 students in 2010 and 12 in 2021. Schools nonetheless remain spread across the territory, under the principle that wherever there is a child there should be a school, and many stay open with only a handful of students (González Pensado et al., 2020; Presidencia de la República, 2026). As a consequence, most rural schools stand alone in the countryside, surrounded only by scattered farmhouses. In our estimation sample, nearly four in five schools had no hamlet or village within 1 km, and for half of these the nearest one was more than 8 km away. The school is also where children eat: in every year from 2003 to 2013, 98 to 99 percent of the rural schools of our analysis sample served meals (ANEP–DIEE, 2021), so the days a child attends school also come with a school meal. Qualitative evidence suggests that this matters: a case study of schools in remote rural areas found food insecurity in half of the households it studied (Ongay Rossi, 2020).

Rural schools also face problems that urban schools face less. Their students attend school intermittently more often, and their teachers are more often new to the school (ANEP, 2010, 2014).[4] Both matter for learning. Additional absences lower achievement in primary school and have lasting effects on later schooling and earnings (Gottfried, 2010; Cattan et al., 2023), and teacher turnover lowers student achievement (Ronfeldt et al., 2013). Rural schools also have less equipment to teach with. In 1996, urban public schools were about four times as likely as rural schools to have a computer and more likely to have a television or a video player, and rural schools not yet connected to the grid had a television or a video player about half as often as connected ones (ANEP, 1996, 2008).[5]

The gap widens after primary school. Nearly all children of primary-school age attended school in both rural and urban areas, but rural youth aged 15 to 17 were about half as likely as urban youth to do so.[6] Primary schools reach the most remote places, but lower-secondary schools do not: there are about ten rural primary schools for each rural lower-secondary program (ANEP, 2014; Sanz Bonino, 2014). Continuing to study therefore means traveling. Rural adolescents are five times as likely as those in larger localities to attend school outside their own locality (Sanz Bonino, 2014), and the lack or remoteness of a school is a reason for dropping out more than three times as often in rural areas and small localities as in the country as a whole (INE–OIT, 2011).[7]

Daily life in the countryside also competes with school. In case studies of rural schools, students travel on horseback, by bicycle, or by motorcycle, and attending is harder where public transport does not reach (Villalba Clavijo, 2016). Rural children are two and a half times as

schools of our estimation sample in 1992–1994.

[4] In ANEP's reports for 2003–2013, rural schools had a larger share of students attending 70 days or fewer than urban schools in eight of the eleven years, and a larger share of teachers with less than three years at the school in every year: between 54 and 66 percent, against about half in urban schools.

[5] Own calculations from the principals' questionnaire of ANEP's 1996 assessment of sixth graders, at the school level: 21 against 5 percent of schools had a computer, 88 against 63 percent a television, and 85 against 59 percent a video player (849 urban public and 201 rural schools). The assessment left out the smallest rural schools, so these gaps are probably understated. Among the rural schools of our analysis sample that took it, 35 percent of the 17 not yet connected to the grid had a television, and as many a video player, against 66 and 63 percent of the 178 already connected. In 2007, 31 percent of rural schools had a computer for administrative use, against 95 percent of common urban schools (ANEP, 2008).

[6] Own calculations from the 1996 census microdata (INE, 1996): 93 against 95 percent at ages 6 to 11, and 29 against 54 percent at ages 15 to 17. Rural is the dispersed rural area, outside INE localities. The 1996 census does not ask directly whether a person attends school; we infer it from the question on whether the person completed the level being attended.

[7] In 2013, ANEP ran 1,107 rural primary schools, and its rural lower-secondary offer amounted to about 110 programs, most of them created after 1986. In 2011, about 55 percent of rural 12- to 14-year-olds who attended school did so outside their locality, against 10 percent in localities of more than 5,000 inhabitants. In 2009–2010, 16 percent of school dropouts aged 5 to 17 in small localities and rural areas cited the lack or remoteness of a school, against 4.5 percent nationwide.

likely as urban children to take part in economic activities (INE–OIT, 2011).[8] Their parents also have less schooling: the mothers of rural children of primary-school age were one and a half times as likely as mothers in the localities outside Montevideo to have no more than primary education.[9] And the countryside has been emptying: between the 1996 and 2023 censuses, the dispersed rural population fell by about two fifths, and nine in ten of the census zones around our schools lost population.[10]

Plan Ceibal, Uruguay's one-laptop-per-child program, gave a laptop to every public primary school student: it started in Florida in 2007, reached the other departments of the interior in 2008, and covered Montevideo and Canelones in 2009 (MEC, 2009; Yanguas, 2020). Using the laptops required electricity to charge them, at school or at home. Our estimates control for the share of each school year in which the laptops had reached the school's department (Section 3).

2.2 The electrification of rural schools, 1992–2021

In Uruguay, the grid is public: the law assigns the public electricity service, the transmission and distribution of electricity to third parties, to UTE, the state-owned electric utility (República Oriental del Uruguay, 1977), so every school on the grid was connected to UTE's network and supplied by it. Even where those who requested a rural extension paid for it, the line joined UTE's network. Our records distinguish the electricity of each school on and off the grid. Off the grid, the electric utility began installing solar panels in rural schools in 1992, and by 2013 nearly one in four of our schools had received one.[11] In 2013 and 2014, Lights to Learn, in collaboration with the Uruguayan government, replaced them with newer systems in the 78 of these schools still off the grid, and UTE took over the maintenance of these systems when the program ended (UTE, 2022; Organización de Estados Iberoamericanos para la Educación, la Ciencia y la Cultura, 2018).[12] By 2015, according to ANEP, no rural school was left without electricity (ANEP–DEMR, 2022). We do not observe other, less reliable sources of power or light, such as generators or kerosene and gas lamps, nor the continuity and quality of supply, so "no electricity" means neither the grid nor a panel.

In 1992, 521 of the 1,149 rural schools open that year were on the grid (Figure 1). Another 56 had a solar panel that year, and 570 had no electricity.[13] By 2005, three in four schools were

[8] In 2009–2010, 14.4 against 5.5 percent of children aged 5 to 14; economic activities include work for the household's own consumption.

[9] Own calculations from the 1996 census microdata (INE, 1996): 72 percent of the children aged 6 to 11 in the dispersed rural area had a mother with primary education or less, against 49 percent in the INE localities outside Montevideo and 35 percent in Montevideo.

[10] Own calculations from INE census microdata (INE, 1996, 2011a, 2026), for persons in private dwellings: the dispersed rural population fell by 37 to 43 percent, from about 8 percent of the country's population to under 5 percent, and 91 percent of the zones lost population. The 1996 census counted people where they were on census night; we keep only those who reported usually living there, because the 2011 and 2023 censuses counted only usual residents. The range reflects two ways of counting the 2023 population: with the INE weight, which adjusts for census omission, or imputing residents to occupied dwellings whose residents were absent, as the 2011 census did; with either count, more than 90 percent of the zones lost population. Part of the national fall between 1996 and 2011 reflects rural places that the 2011 census counts as localities. The zones are the dispersed rural parts of the census areas around the schools of our analysis sample, followed over approximately the same territory in the three censuses.

[11] 276 of the 1,159 schools of our analysis sample (DNETN–MIEM, 2009; UTE, 2024a; ANEP–DEMR, 2022).

[12] These records date grid connections and panels, some only by a bound or a range of years (Section 2.3). The OEI counts 79 schools with the program in Uruguay (Organización de Estados Iberoamericanos para la Educación, la Ciencia y la Cultura, 2018).

[13] Section 2.3 describes how we date each connection and each panel. UTE's electronic records begin on 1 May 1992 and date most earlier connections to that day, so for 503 of the 521 schools on the grid in 1992 we know only that they were connected by then. All 56 panels of 1992 are dated only by the range 1992–1995, and we count them from its first year, so 56 is an upper bound and 570 a lower bound. Two other schools open in 1992

on the grid, and most of the rest had a solar panel. Of the 1,039 schools open in 2021 whose source we know, the 15 off the grid had Lights to Learn panels.[14]

Our records show that the grid reached schools more than 5 kilometers from any locality of the 2011 census later than the rest (Appendix Figure 8). In 1995, it reached 88 percent of village schools but 23 percent of the schools more than 10 kilometers from any census locality.[15] Solar panels went mostly to schools more than 5 kilometers from any census locality: 232 of the 276 schools that had one by 2013 (Appendix Figure 8).

Counting only the years with a known source, the schools followed five routes between 1992 and 2021 (Figure 2): 352 went from no electricity to the grid, 220 from a solar panel to the grid, and the rest were always on the grid, had a panel but never the grid, or never had electricity. The two routes to the grid came in different periods: 80 percent of the connections from no electricity took place by 2001, and over 90 percent of those from a solar panel after 2002. Schools already on the grid in 1992 count as on the grid throughout, so the variation in the timing of electrification comes from the schools that changed source.[16] A panel already supplied electricity, so Section 3 estimates each change of source separately: from no electricity to the grid, to a panel from the electric utility, or to either; from such a panel to Lights to Learn; and from a panel to the grid.

2.3 Data

We build a panel of all rural public primary schools in Uruguay, observed every year from 1992 to 2021. Its core is ANEP's administrative records of public primary schools (ANEP–DIEE, 2021), which report for each school and year its enrollment at the end of the school year, the number of students in each band of days attended, repetition and, until 2015, the number of teachers. We complement them with the 2005, 2010, 2015, and 2020 waves of ANEP's survey of students' households, which describe the socioeconomic context of each school. The panel covers the universe of rural schools ($N = 1{,}380$) over a span of three decades.[17]

We date each school's connection to the grid with the electric utility's records of the electricity supplies of ANEP's schools, obtained through a public-information request (UTE, 2024c), and its solar panels with the utility's report of the panels it installed in rural schools (UTE, 2024a).[18] Combining these dates with documents that show a school still without the grid at a

have an undetermined state. In Figure 1, a school without a connection date counts as on the grid from an upper bound on it; for 38 of the 40 such schools, the bound is ANEP's statement that from 2015 every rural school outside Lights to Learn was on the grid.

[14]The other 5 schools open in 2021 are those whose lack of grid until 1996 we infer from that year's census; no record of these schools gives their source after 1996, and we leave those years out, also in Figure 1; all 5 stayed open through 2021.

[15]Among the schools with a connection date after 1992, the median year of connection was 2004 beyond 10 kilometers, against 1997 to 1999 within 5 kilometers.

[16]These are the 504 schools connected by 1 May 1992, 503 of them dated to that very day; each counts as on the grid in every year it was open.

[17]Each school enters in every year it appears in ANEP's records, so that a change of category does not remove a school from the panel. Some changes were temporary: between 2000 and 2005, ANEP classified 92 of these schools as urban for some years and then returned them to the rural category.

[18]For each supply, the records give the date of the first service ever provided at the school's premises, which we take as the connection date. When sources disagree, the connection year in the electric utility's 2022 report on the schools of Lights to Learn (solar panels) takes precedence, followed by the records of ANEP's Department of Rural Education and by the utility's 2024 re-verification of 128 schools that its first delivery had dated after 2013 (UTE, 2022; ANEP–DEMR, 2022). The records of ANEP's Department of Rural Education complete the panels of four schools, and the utility's 2022 report gives those of Lights to Learn (ANEP–DEMR, 2022; UTE, 2022). Two features of these records limit the precision of the dates. UTE's electronic records begin on 1 May 1992, and supplies connected earlier generally carry that date: of the 1,068 schools of the analysis sample with a connection date, 503 carry it, so we know only that they were on the grid by then. UTE also dates 180 of the

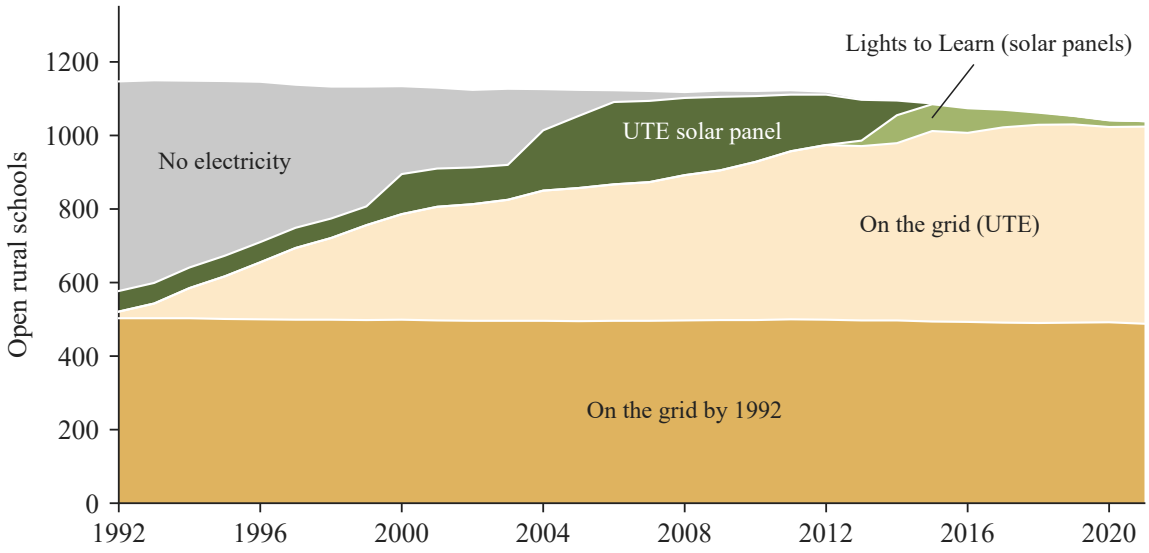


Figure 1: Rural schools by source of electricity, 1992–2021.

Notes: Rural primary schools of the analysis sample, 1992–2021 (N = 1,159 schools, 33,450 school-years). The figure leaves out the years after 1996 of the 5 schools whose lack of grid until 1996 we infer from the 1996 census. A school counts as on the grid from its connection year or, for 40 schools without a connection date, from an upper bound on it. The darker yellow area, labeled On the grid by 1992, marks the schools that UTE dates to 1 May 1992, when its electronic records begin, which means connected on or before that day. A panel counts from its installation year or, when UTE gives only a range of years, from the first year of the range. Not shown: 56 school-years with an undetermined state. Source: own elaboration based on ANEP and UTE records.

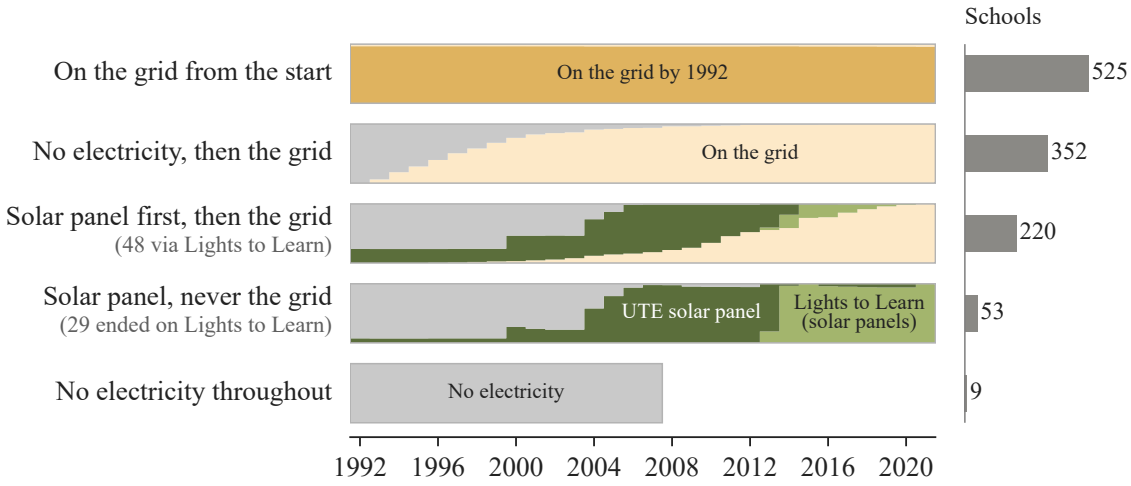


Figure 2: Source of electricity of rural schools, year by year, by route to the grid, 1992–2021.

Notes: Each row is a route: the sequence of sources a school had while open (N = 1,159 schools of the analysis sample). Its strip gives, each year, the share of the route's open schools on each source, in years with a known source, and ends in the route's last year with an open school. The bars count the schools per route. Of the 220 schools that went from a solar panel to the grid, 37 have only an upper bound on the connection date and 28 only its year. Source: own elaboration based on ANEP and UTE records.

given date, we classify every year in which a school was open as on the grid, with a solar panel, or without electricity.[19]

We locate 1,179 schools at their own coordinates.[20] These points give each school its distance to the localities of the 2011 population census, which measure isolation (Section 2.1), and to those of the 1996 census, which we use whenever a measure must predate most connections, as in the index of rurality below and in Appendix Table 5 (INE, 1996, 2011a,b). The coordinates also place most schools in a rural census zone, so the 1996, 2011, and 2023 censuses tell us how many people and dwellings surrounded them (INE, 2026). We call a school isolated if the nearest populated place delimited by the 2011 census, of any size, from a hamlet of a few dwellings to a city, lies more than 1 km away; otherwise it is next to a hamlet, with fewer than 300 inhabitants, or a village, with 300 or more.[21] Population within 1 km comes from GHSL for 1990 (Schiavina et al., 2023b): in our estimation sample, the median isolated school had about 9 people living within 1 km, against about 170 for schools next to a hamlet or a village. Gridded population and settlement classes from GHSL, building footprints from Open Buildings, and night lights from DMSP (1992–2013) and VIIRS (since 2014) describe the people, buildings, and lights around each school; the lights help us follow the spread of the grid beyond the school.[22] Daily rainfall from CHIRPS, unusually cold and unusually hot school days from NASA POWER, and river discharge from GloFAS measure the weather of each school year, which our attendance estimates control for.[23] Weather matters more for attendance in the countryside than in cities: rural students often reach school on horseback, by bicycle, or by motorcycle, where public transport does not reach (Section 2.1), so rain, floods, and unusually cold or hot days can keep them at home more easily. The coordinates also place each school in one of 217 police sections, the unit at which we cluster standard errors,[24] and its department places it in one of the six regions of the national emergency system (SINAE), within which we compare schools (Section 3).[25]

272 panels in its report only by a range of years. We count each of them from the first year of its range, except one that the minutes of a departmental legislature date earlier.

[19]These documents include UTE's 2019 departmental reports on rural electrification, UTE's list of the rural schools without the grid that the 2009 agreement between ANEP and UTE covered, minutes of departmental legislatures, and UTE news releases (UTE, 2019, 2024b). A 2022 note of ANEP's Department of Rural Education adds that, from 2015, no rural school was left without electricity (ANEP–DEMR, 2022). For 33 schools, the connection date itself comes from other records of the same school, mostly other UTE records of its supplies.

[20]Nearly all come from ANEP's 2024 geographic layer of primary schools, which places 1,103 of the 1,179 (ANEP, 2024). The other 76 come from ANEP's register of schools, from maps of closed schools prepared in an earlier stage of this project, or from the hamlet each school reports, and three were corrected after a check against their police section (GeoNames, 2026). We locate 173 more schools at points from secondary sources that we validate against the schools whose location we know. Most of these points come from the school symbols on departmental maps published online in 2012, and the rest from online map listings, OpenStreetMap, or the hamlet each school reports (notocactus.eu, 2012; OpenStreetMap contributors, 2026). Of the 40 such schools in the analysis sample, 26 have a point that one or two sources place at the school itself. The 28 schools located only by their department have no location-based data and are not in the analysis sample.

[21]We measure the distance to the edge of the locality. We use the localities of the 2011 census because they are more disaggregated: the 2011 census delimits many small localities that the 1996 census did not, so that 280 schools of our analysis sample have a locality within 1 km with the 2011 localities and 195 with those of 1996, 193 of them with both. That the 2011 classification postdates most connections is unlikely to reflect settlement attracted by electricity: between the 1996 and 2011 censuses, population fell in 93 percent of the census zones around our schools.

[22]Schiavina et al. (2023b,a); Sirko et al. (2021); Elvidge et al. (1997, 2013); Appendix Table 3 lists every source with its period and coverage.

[23]The sources are Funk et al. (2015), NASA Langley Research Center (2026), and Zsoter (2019). A cold (hot) day is a school day whose minimum (maximum) temperature falls among the coldest (hottest) 5 percent of the days of 1992–2021 in the school's area; the thresholds are relative to each area, not absolute.

[24]The 1,102 schools of the estimation sample lie in 206 police sections; each of the 2 schools without one is its own cluster. The police sections come from INE (2010) and the regions from SINAE (n.d.).

[25]The size of the rural parcels in the current cadastre and the distance to the four hydroelectric dams are the candidate instruments that Appendix B tests (DNC, 2026; MIEM, 2026).

Our analysis sample keeps 1,159 of these schools, observed in 33,450 school-years between 1992 and 2021. We drop 221 schools, 214 of them because we have neither a usable grid date for them nor a record of any year in which they lacked electricity.[26] ANEP's rural label is an administrative category that does not always reflect how rural a school is: between 2000 and 2005, ANEP moved 92 of the schools in our panel to the urban category for some years and then back, and the category includes some schools next to localities of 5,000 inhabitants or more. We therefore rank the 1,159 schools with an index of rurality that combines their size and isolation in the first decade of the period.[27] Our broad sample drops the 5 percent least rural schools and keeps 1,102. Our reduced sample drops the 15 percent least rural and keeps 990.

Our main outcome is the share of students in grades 1 to 6 who reach satisfactory attendance in the school year, which in Uruguay runs from March to December, within the calendar year. ANEP's records count the students of each school by bands of days attended, not student by student (ANEP, 2010): 70 days or fewer, which ANEP calls intermittent dropout; 71 to 140 days, which it calls insufficient attendance; and 141 days or more, which we call satisfactory. We measure each band as a share of final enrollment in grades 1 to 6, so the three shares add up to one in every school-year.[28]

We also study three other outcomes: repetition, the share of these students who did not pass to the next grade; enrollment in grades 1 to 6 at the end of the school year; and the number of teachers of those grades, which, in schools this small, determines how many grades share a classroom, since three in five had a single teacher for grades 1 to 6. Enrollment covers 1992–2021 and the number of teachers 1992–2015.[29] We end the attendance and repetition series in 2019, before the pandemic: across the schools of the analysis sample, the mean share of students attending 141 days or more fell from 88 percent in 2019 to 6 percent in 2020. Appendix Table 1 describes these outcomes in 1992–1994, and Appendix Tables 2 and 3 list every source with its period and coverage. A second set of outcomes comes from the 1996 and 2011 population censuses: the share of young people aged 18 to 24 in the census zone around each school who completed lower secondary school (Section 4).[30]

Appendix Table 1 describes our estimation sample, the broad sample of 1,102 schools, in 1992–1994. The median school had 15 students in grades 1 to 6, three in five had a single teacher, and nearly four in five were isolated, with the nearest locality of 5,000 inhabitants or more a median of 23 km away. Only 43 percent were on the grid in 1992, and on average 84 percent of their students reached satisfactory attendance.

[26] Three of the other seven schools had no students in any year and neither a grid date nor a bound on it. One has a grid date that contradicts its inclusion in UTE's list of the schools without the grid that the 2009 agreement covered (UTE, 2024b). Three have a grid date that both the 1996 census and the principal and household questionnaires of ANEP's 1996 assessment of sixth graders contradict (INE, 1996; ANEP, 1996).

[27] The index averages five standardized measures, each in logs and signed so that higher values are more rural. Two measure size: total enrollment and teachers, averaged over 1992–2002. Two measure isolation in the 1996 census: the distances to the nearest localities of at least 1,000 and at least 5,000 inhabitants. The fifth is the population within 5 km of the school, averaged over 1992–2002, from GHSL (Schiavina et al., 2023b). We standardize the measures among the 1,119 schools of the sample with their own coordinates. The 5 percent least rural schools lie a median of 1.1 km from a locality of 5,000 inhabitants or more in the 1996 census; the median for the broad sample is 24.4 km.

[28] A 2004 resolution of the primary education council required students to attend at least 80 percent of school days to be promoted, and allowed inspectors to approve exceptions; its 2008 protocol on absenteeism equated ANEP's threshold of insufficient attendance, 140 days, with about 76 percent of the days a student should attend (ANEP–CEP, 2008). ANEP's records count students only by these bands, so 141 days is the closest threshold to the promotion rule that the data allow.

[29] ANEP's records have no count of the teachers of grades 1 to 6 for 2003, and the counts for 2002 repeat those of 2001 in all 1,232 schools with both years, so we treat 2002 as missing.

[30] National learning assessments cannot serve as an outcome: the 1996 assessment of sixth graders reached only 16 of the 436 schools without electricity in our sample, and the later waves use different tests.

3 Empirical strategy

Our setting has two features that rule out a standard two-way fixed effects regression. A school's electricity takes one of three states (no electricity, a solar panel, or the grid), and a school can move through several of them, always toward more electricity: from no electricity to a solar panel or to the grid, and from a solar panel to a newer one or to the grid (Figure 2); and the effect of a connection may differ across cohorts and over time. In such settings, a two-way fixed effects regression compares switching schools with schools that switched earlier and can give some of these comparisons negative weights (de Chaisemartin and D'Haultfœuille, 2020). We therefore estimate each change of source separately with the difference-in-differences estimator of de Chaisemartin and D'Haultfœuille (2026). For each change, the estimator compares the schools that make it in a given year with schools in the same region that started the period in the same state and had not yet changed; a school with a solar panel, for instance, never serves as a control for a school that goes from no electricity to the grid. Our main design is the move from no electricity to the grid; we also estimate the moves from no electricity to a solar panel and to either source, and, in the appendix, the moves from a utility panel to Lights to Learn and from a panel to the grid.

Formally, let $Y_{s,t}$ be the outcome of school s in year t and F_s the first year in which its electricity status differs from its status in 1992. For a school that switches, the effect ℓ years after the switch is estimated by

$$\mathrm{DID}_{s,\ell} = (Y_{s,F_s-1+\ell} - Y_{s,F_s-1}) - \frac{\sum_{s' \in \mathcal{C}_{s,\ell}} w_{s'} \left(Y_{s',F_s-1+\ell} - Y_{s',F_s-1}\right)}{\sum_{s' \in \mathcal{C}_{s,\ell}} w_{s'}}, \tag{1}$$

where $\mathcal{C}_{s,\ell}$ is the set of schools in the same region as s that had the same status in 1992 and had not yet switched by year $F_s - 1 + \ell$, and w_s is the school's enrollment in grades 1 to 6 in its first year for attendance and repetition, and 1 for the other outcomes. The event-study estimate DID_ℓ averages $\mathrm{DID}_{s,\ell}$ across the switching schools with the same weights, for $\ell = 1, \ldots, 6$. Our headline estimate is the average of the first two, $\widehat{\beta} = (\mathrm{DID}_1 + \mathrm{DID}_2)/2$, for attendance and repetition, and of the first five for enrollment, teachers, and closure. We fixed these summaries before estimating: the first school year with electricity is often only partly treated, so the first two years capture the immediate effect, while enrollment and teachers adjust more slowly. Placebo estimators replace $Y_{s,F_s-1+\ell}$ with $Y_{s,F_s-1-\ell}$, for $\ell = 1, \ldots, 4$, and compare switching and control schools before the switch.

We specified the timing of treatment before estimating any effect. For attendance and repetition, which are measured over the whole school year, a connection counts from the school year in which it occurred, except for connections in November or December, the last two months of a school year that runs from March to December, which count from the next school year. For enrollment, teachers, and school closure, which are determined at the start of the school year, it counts from the following year.

Our attendance estimates control for the weather of each school year (school days with river floods, rainfall, and unusually cold and hot days) and for the share of the school year in which Plan Ceibal's laptops had reached the school's department; the estimates for the other outcomes control only for Plan Ceibal. As in de Chaisemartin and D'Haultfœuille (2026), each change in the outcome is residualized on the changes in the controls, with coefficients estimated among the schools that had not yet switched, and we report every estimate also without controls. Standard errors are clustered by police section. We also report p-values from a pairs bootstrap that resamples police sections and reestimates every step, including the coefficients of the controls (999 replications), and from a permutation of connection dates that preserves how they cluster within police sections (500 permutations).

The estimator rests on two assumptions: that schools did not change their outcomes in anticipation of a connection, and that, absent a change of source, switching schools and their controls would have followed parallel paths within each region. The grid did not reach schools at random: it reached first the schools closer to the existing network and to lit areas, in more populated places, and with more students (Appendix Table 5). Our design does not require random timing, only that these differences in levels did not translate into differences in trends. We assess this in two ways. First, the placebo estimators test whether switching and control schools followed parallel paths in the four years before the switch; anticipation would show up in the placebo for the year before it. Second, we let the trends of each school vary with its initial characteristics (its isolation, the population around it, and its enrollment in 1992) and with the socioeconomic context of its census zone in 1996 (Section 4).

We also explored whether a predetermined feature of each school's location could serve as an instrument for the timing of its connection. Following the literature that uses geography to instrument electrification (Dinkelman, 2011; Lipscomb et al., 2013), we tested two candidates: the distance to the four hydroelectric dams, built between 1945 and 1982 where hydrology dictated, and the size of the largest rural parcel around each school, because rural extensions of the grid were often paid for by those who requested them. Neither works (Appendix B). Within region, both instruments have weak first stages, with effective F statistics of at most 7.6 and 2.4, respectively, against a critical value of 23.1 (and at most 4.4 for the dams after excluding schools within 100 km). Moreover, the Anderson–Rubin confidence sets are either uninformative or imply implausibly large effects of electrification. While these findings raise concerns about the exclusion restriction, they do not formally establish its violation.

4 Results

Connecting a school to the grid raised the share of its students who reached satisfactory attendance (Appendix Table 7, column 1). Over its first two school years with electricity, the share rose by 3.95 percentage points (standard error 1.23), against an average of 84 percent in school-years without electricity. The estimate rests on 337 schools that moved from no electricity to the grid, and it holds under each form of inference: the pairs bootstrap gives a 95 percent confidence interval from 0.98 to 6.80 points ($p = 0.008$), and the permutation of connection dates gives $p = 0.004$. The four placebo estimates are jointly insignificant ($p = 0.60$), consistent with switching and control schools following parallel paths before the connection (Figure 3). In the third to sixth school years with electricity, the estimates stay positive, between 1.8 and 3.0 points, but they are less precise, and none is significant at the 5 percent level. In the reduced sample, which drops the 15 percent least rural schools, the gain is 4.36 points (1.32).

Because the three attendance shares add up to one in every school-year, the bands show which students electricity moved. Almost all of the gain came from students who would otherwise have attended irregularly: the share with insufficient attendance (71 to 140 days) fell by 3.93 percentage points (1.05), and its placebos also show no differences before the connection ($p = 0.65$). The share in intermittent dropout, about 2 percent of students, did not change (-0.01, standard error 0.46). Electricity thus moved irregular attenders up to satisfactory attendance rather than bringing back students who had nearly stopped attending. For a school of the median size, with 15 students, the gain amounts to about 0.6 students per school year. The attendance gain is consistent across grades (the difference between grades 1–3 and 4–6 is 1.2 percentage points, standard error 1.6).

Pooling the two sources, moving from no electricity to either one raised satisfactory attendance by 3.45 points (1.19; Appendix Table 7, column 2), an estimate driven by the grid: nearly

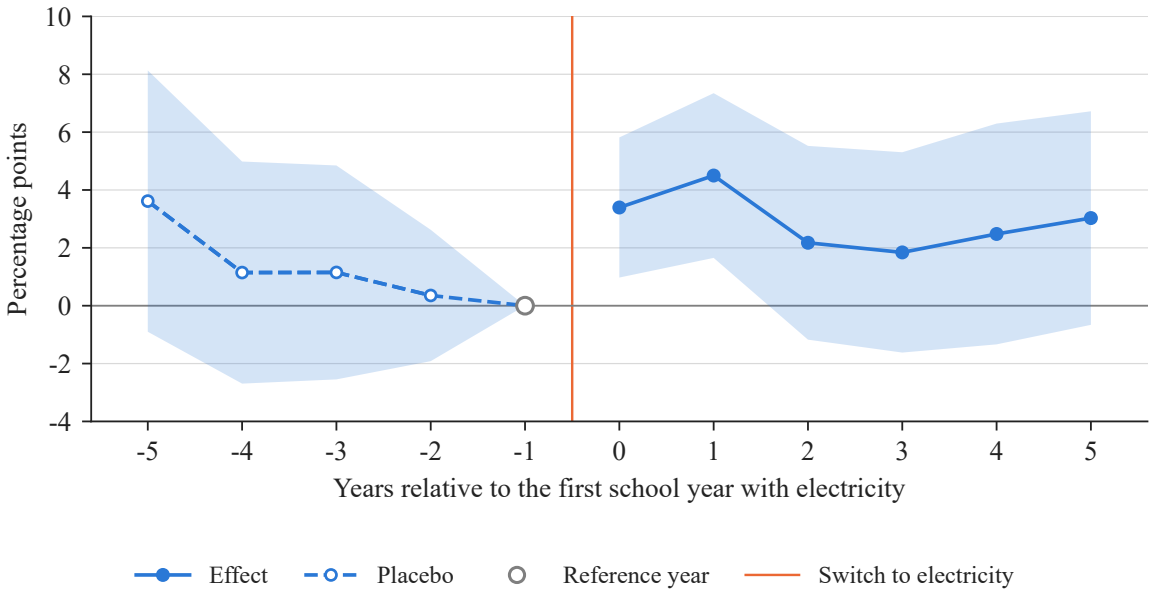


Figure 3: Effect of a grid connection on satisfactory attendance, by year since the connection.

Notes: Satisfactory attendance: share of students in grades 1–6 attending 141 days or more. Solid line: effects of connecting a school without electricity to the grid, from year 0, its first school year with electricity, to year 5. Dashed line: placebos, the outcome in each earlier year minus that in year −1 (the reference, set to zero), relative to schools not yet connected; a positive placebo means a relatively higher outcome before. Shaded bands: 95 percent confidence intervals, clustered by police section. Specification and sample as in Appendix Table 7, column (1). Source: own elaboration based on ANEP and UTE records.

four in five of its 431 switching schools were connected to the grid. Solar panels alone produced no detectable gain: 1.94 points (3.55), estimated on 93 schools (column 3). The estimate is too imprecise to tell whether panels had no effect or one as large as the grid's: its 95 percent confidence interval runs from −5.0 to 8.9 points. Two other changes of source cannot be read as effects (Appendix Table 12). Moving from a solar panel to the grid gives 1.16 points (2.20), but its placebo estimates reject parallel paths before the change. Replacing a utility panel with a Lights to Learn panel gives 18.9 points (11.7), but the estimate rests on 48 schools and, in its second year, on only two comparison schools, so it is too imprecise to interpret.[31] Counting as UTE-panel years those of unknown grid status, a variant chosen after seeing its result, 73 of the 78 schools reached by Lights to Learn enter the first-year comparison, and the estimated gain in satisfactory attendance is 10.2 percentage points (SE 5.0; Appendix Table 12). The estimate is fragile: in the second year, 39 of the 48 schools followed are still compared with the same two control schools, in a single police section, and the pairs bootstrap, which leaves that section out in about a third of its draws, gives a standard error of 13.9 ($p = 0.46$); the permutation p-value is 0.05.

The size of the gain is less robust than its sign. Dropping the enrollment weights, the 1994 cohort (the one that weighs most), or the cohorts connected before 1997, or keeping only the

[31] Lights to Learn reached 78 schools, but 48 enter the estimate. Twenty-five had no school in their region documented as still running on the utility's panel without the grid in the same years: 24 of them had their first school year with the new panel in 2014, when only five schools, all in Tacuarembó, met that condition. Two no longer appear in ANEP's records by their first school year with the new panel, two lack attendance records for the year before or the year of the change, and one had no earlier utility panel. Forty-two of the 48 are in the Northeast, and the 40 followed into the second year are compared with only two schools. Counting panel years of unknown grid status as UTE-panel years, 73 schools enter the first-year comparison, but 39 of the 48 followed into the second year are still compared with those two schools (Appendix Table 12).

connection dates of high certainty, lowers it to between 2.4 and 3.0 points, and not all of these estimates are significant at the 5 percent level.[32] Other changes leave it close to 4 points. Letting schools follow different trends by their initial characteristics, by the socioeconomic context of their census zone in 1996, or by region and isolation keeps it between 3.51 and 4.55 (Appendix Table 10), and dropping the controls, comparing schools within department rather than region, or allowing for a year of anticipation gives 3.50 to 4.10 (Appendix Table 9).

The gain did not wait for Plan Ceibal's laptops. It rests on the connections of the 1990s, which are most of the switching schools: for the 285 schools connected between 1993 and 2002, out of the 337 that switched, the gain is 3.52 points (1.31), years before laptops reached rural classrooms in 2008. Later connections are too few to tell whether their effect was smaller: for the 52 schools connected from 2003 on, the estimate is 0.34 points, with a 95 percent confidence interval from -6.1 to 6.7 that contains the gain of the 1990s. Several changes would lead one to expect a smaller effect later, although we cannot test it. From 2003 on, nearly all the schools connected to the grid already had a solar panel (Appendix Figure 10); households off the grid increasingly had other sources of electricity: around isolated schools, the share of dwellings without the grid that had one rose from 37 percent in 1996 to 66 percent in 2011, mostly through solar chargers; and satisfactory attendance had already risen from 86 percent in 1992 to about 90 percent or more from 1997 on. Nor did the laptops themselves detectably raise attendance. In schools on the grid throughout 2001–2012, satisfactory attendance did not change when the laptops reached their department, relative to schools still off the grid (-0.15 points, standard error 1.45), and across 2,394 specifications of the interaction between electricity and the laptops' arrival, none survives a correction for multiple testing.[33]

Repetition fell by 1.26 percentage points (standard error 0.74; Appendix Table 7), but we do not read this as an effect: the placebo estimates reject parallel trends before the connection ($p = 0.01$), and the permutation of connection dates gives $p = 0.11$.[34]

Enrollment did not detectably respond (Appendix Table 11). Over the first five school years with electricity, enrollment in grades 1 to 6 rose by 2.6 percent (standard error 2.3), and the 95 percent confidence interval rules out increases larger than about 7 percent. There was little room for it to grow: in 1996, 93 percent of rural children aged 6 to 11 already attended school (Section 2.1), so the gain came from children who were already enrolled attending more days rather than from new students.

A connection could still have drawn students away from nearby schools without electricity. Figure 4 follows the 29 pairs in which a school was connected to the grid and a neighbor within 5 km was not. The neighbor's enrollment fell after the connection, but by about as much as that of the neighbors 20 to 50 km away, which reflects the general decline of rural enrollment: relative to them, the neighbor lost 4.8 log points one year after the connection, with a 95 percent confidence interval from -17.5 to 5.2, and no neighbor closed within three years. The connected school's enrollment rose, but it was already growing faster than that of the distant schools before the connection, so the rise cannot be attributed to the grid. With so few close pairs we cannot rule out a moderate shift of students, but we find no evidence of one.

Electricity did not change staffing either (Appendix Table 11). Over the first five school

[32] Without enrollment weights, 2.89 (1.27); without the 1994 cohort, 2.38 (1.25); with only the cohorts connected from 1997 on, for which all four placebos are observed, 2.96 (1.46); with only the connection dates of high certainty, 2.95 (1.64).

[33] The first comparison, of 812 schools on the grid with 27 without it, also shows breaks in years without new laptops (2006 and 2010), so it is at most suggestive. The specifications of the second vary the design, sample, window, and timing of the laptops' arrival; the lowest p-value corrected for the whole search is 0.41.

[34] A rule of the primary education council approved in November 1999 (Circular 441) restricted repetition in first grade to exceptional cases. Splitting the years around it, the clearest signal is a fall of 4.4 to 5.6 points in first-grade repetition before the rule, which does not reach the 5 percent level either.

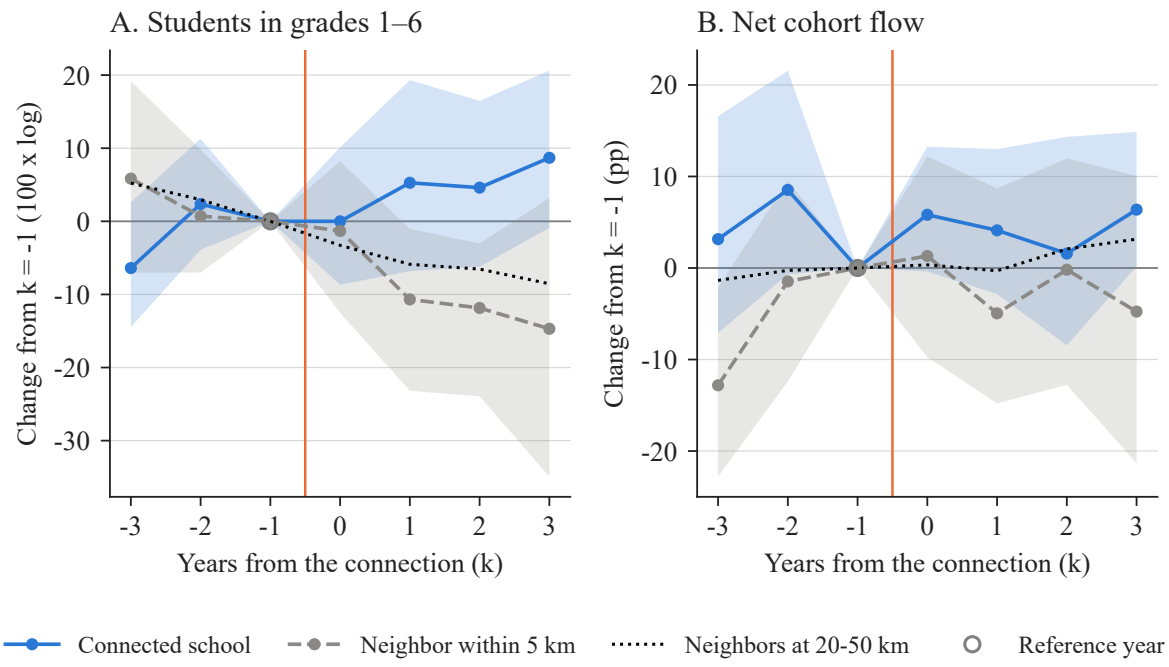


Figure 4: Enrollment of a newly connected school and of its close neighbors without the grid.

Notes: Rural schools connected to the grid (28 events) and their neighbors without the grid within 5 km (29 pairs); $k = 0$ is the first school year with the grid, and changes are relative to $k = -1$. Panel A: students in grades 1–6, 100 × log. Panel B: net cohort flow (students in grades 2–6 over those in grades 1–5 the year before), in percentage points. Dotted line: the neighbors of the same events at 20 to 50 km, the reference for the general fall in rural enrollment. Bands: 95 percent intervals from 1,000 bootstrap draws of the events. Relative to the dotted line, the neighbor's fall is not significant, and the connected school was already growing before the connection. Source: own elaboration based on ANEP and UTE records.

years with electricity, the number of teachers of grades 1 to 6 did not move (−0.00, standard error 0.02), which rules out changes larger than about 0.04 teachers per school.[35] Nor were connected schools less likely to close (0.63 percentage points, standard error 0.48), although this comparison is only descriptive: schools about to close may never have been connected, and a school must be open to be connected, so its placebos reject by construction.

ANEP's records do not report attendance by sex, so we cannot study gender differences in our main outcome. For repetition and enrollment, records by sex begin only in 2004, and the few connections after that year leave the differences between girls and boys too imprecise to interpret.[36] In the censuses, school attendance does not change detectably for either girls or boys.

The 1996 national assessment of sixth graders offers a glimpse of learning. Students in the 27 rural schools connected to the grid between 1992 and 1996 scored 0.04 to 0.19 standard deviations higher in reading and 0.05 to 0.27 higher in math than those in the 16 schools that still had no electricity in 1996, depending on the controls (Appendix Table 13).[37] All of the differences are positive, but none is statistically significant at the 5 percent level: with 43 schools, the comparison can detect only differences of about half a standard deviation or more, so it does not support a causal reading.

The school records follow children only while they attend a rural primary school, so they cannot tell whether more of them went on to secondary school. The 1996 and 2011 population censuses can, at the level of the census zone around the schools. We compare the 137 zones whose first school was connected to the grid between the two censuses with the 42 zones whose schools were still off the grid in 2011, nearly all of them with a solar panel.[38] Among young adults aged 18 to 24, the share who had completed lower secondary school rose from 18 to 40 percent in the connected zones and from 15 to 25 percent in the others, a difference of 10 percentage points (standard error 4.4) once we control for the zones' characteristics in 1996 (Figure 5). Over the same period, the share of households on the grid rose 31 percentage points more in the connected zones, so the estimate captures the electrification of the zone, not the school's connection alone. We read it as suggestive: with a single census before the connections we cannot test for diverging trends, and the census records where people live, not where they grew up.[39]

[35] Teacher experience and tenure in the school are recorded only from 2002, so only the 46 to 49 schools connected from 2003 on inform them, too few to detect less than about 20 percentage points.

[36] Between 31 and 51 schools switch after 2004. Repetition fell by 5.4 percentage points among boys (standard error 4.0) and did not change among girls (1.1; $p = 0.14$ for the difference), and enrollment rose by 9.8 percent among girls and fell by 1.6 percent among boys ($p = 0.30$ for the difference).

[37] The controls enter in turn: the school's sixth-grade outcomes in 1992–1994, household characteristics from the assessment questionnaires, and region fixed effects. The only difference close to the 10 percent level is the one in math with region fixed effects ($p = 0.090$ with the wild bootstrap and 0.115 with permutation inference). In the nine schools connected in 1995–1996, the reading difference is 0.35 to 0.46 standard deviations; it is significant at the 5 percent level only without controls and not once household characteristics are added.

[38] Zones are the territorial units that cover the same ground in both censuses, outside Montevideo, and contain at least one rural school. Each outcome is a separate regression of the zone's 1996–2011 change on an indicator for connected zones, weighted by the zone's 2011 population in the age group and controlling for the outcome's 1996 level, the 1996 share of households on the grid, log population in 1996, log number of schools, the distances to the nearest locality, to a city of 5,000 or more, and to a hydroelectric dam, population within 1 km in 1990, built-up area within 1 km in 1995, and night lights within 5 km in 1992–1995. In 39 of the 42 comparison zones a school had a solar panel.

[39] Unweighted, the difference is 11.3 percentage points (standard error 5.5). In 1996, lower secondary completion was the same in both groups of zones (a difference of 0.4 points, $p = 0.91$). Its Benjamini–Hochberg q-value across the nine census outcomes we estimated is 0.11. Among people aged 25 to 34, most of them adults when the grid arrived, years of schooling rose 0.27 years more in connected zones ($p = 0.39$; unweighted, 0.62 and $p = 0.03$), which suggests that part of the difference may reflect who remained in each zone.

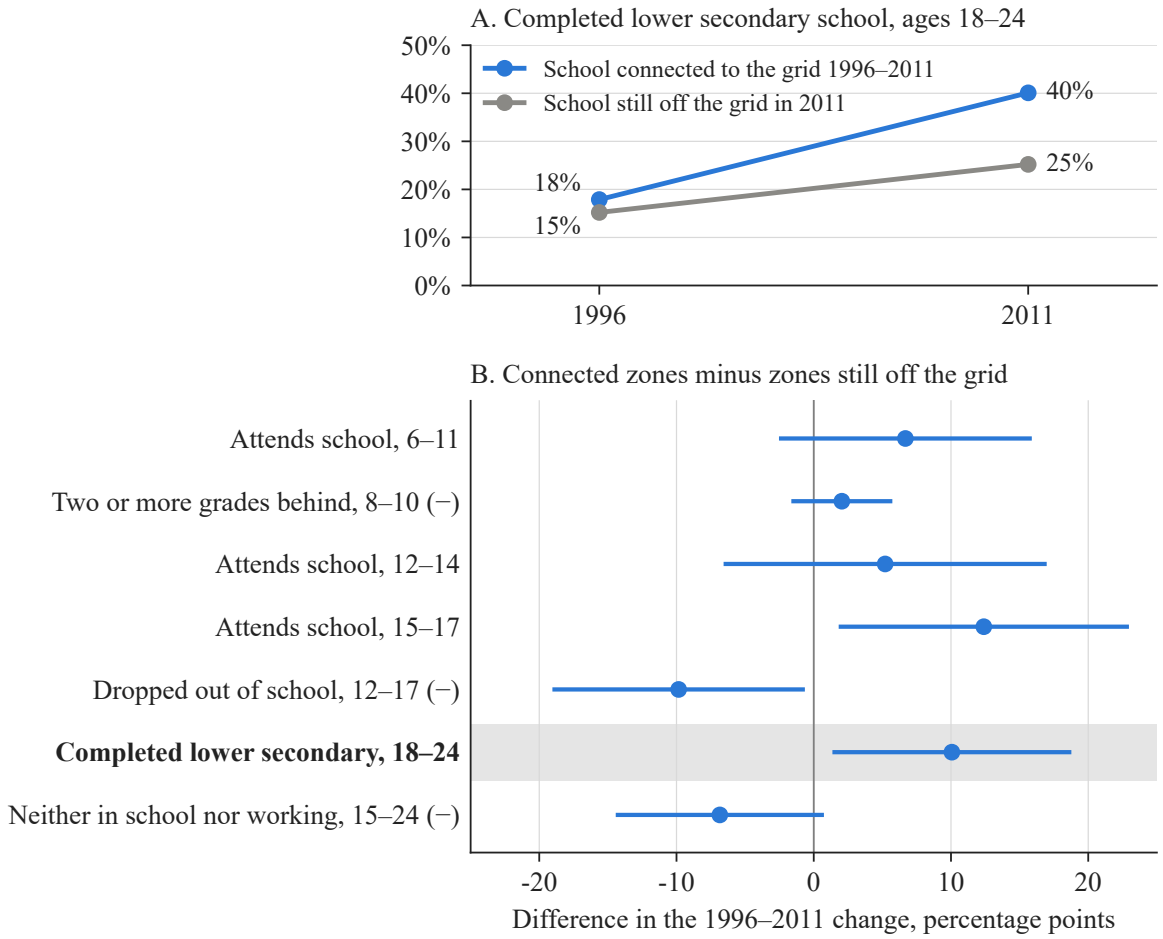


Figure 5: Completion of lower secondary school and other schooling outcomes in the 1996 and 2011 population censuses.

Notes: Census zones with a rural school; connected zones: first school connected to the grid between the censuses (137 zones, 226 schools); comparison zones: first school still off the grid in 2011 (42 zones, 54 schools). Panel A: share of all people aged 18 to 24 in the zones who completed lower secondary school (nine or more years), in the 126 connected and 39 comparison zones with people of that age in both censuses. Panel B: difference between connected and comparison zones in the 1996–2011 change of each outcome, in percentage points, from separate regressions with the controls listed in the text, weighted by the zone's 2011 population in the age group; 151 to 173 zones per outcome. Bars: 95 percent confidence intervals from heteroskedasticity-robust standard errors. (−): a fall is an improvement. Source: own elaboration based on INE's 1996 and 2011 censuses and ANEP and UTE records.

5 Mechanisms

Electricity could have raised attendance by drawing children away from work. If so, the gain should be larger where more children worked. We detect no such difference: in the census zones where more children aged 12 to 14 worked in 1996, satisfactory attendance rose by 1.6 percentage points (standard error 2.3) after a grid connection, against 5.9 (2.6) where fewer did.[40]

Two other channels predict different patterns. In the first, the connected school becomes the only lit place around, and a more attractive one. Classrooms with light, and later with appliances, can draw children who would otherwise stay home, and a better place to work can help a remote school attract and retain teachers (United Nations Department of Economic and Social Affairs, 2014), although the number of teachers did not change in our schools. Part of such a gain may fade as the novelty wears off, as the attendance gain from solar lanterns did in rural Bangladesh (Kudo et al., 2019). This channel predicts the largest gains where the households around the school stayed in the dark. In the second, the grid also reaches those households, where children can study after dark and use the laptops that Plan Ceibal gave every public primary school student (Yanguas, 2020). This channel predicts the largest gains next to hamlets, where the grid could follow the school to nearby dwellings.

The gain is larger, and lasts longer, far from larger localities. In schools farther than the median, 23 km, from a locality of 5,000 inhabitants or more, satisfactory attendance rose by 6.6 percentage points (standard error 1.9) over the first two school years with the grid, and the yearly estimates stay between 4.5 and 7.5 points over the first six. Closer to such a locality, the gain, 1.1 points (1.8), cannot be told apart from zero. The difference between the two groups is imprecise, however: it does not survive the pairs bootstrap or a correction for the eleven splits we tested (Appendix Table 14).

Next to a hamlet, the grid followed the school to the households, but how soon it did is far from obvious. In the censuses, dwellings in hamlets next to a school had almost no grid electricity before the school was connected and nearly all had it afterward (Figure 6). With censuses 12 to 15 years apart, and few schools observed in a census taken just before or just after their connection, this says only that the dwellings were connected at about the same time as the school. Monthly night lights, available for a few schools connected in 2012–2019, show that the lag varies widely (Appendix Figure 12): the light of one hamlet rose the month after the school's connection; that of two others rose only modestly, four and five months after it, and in one of them most of the rise came almost six years later; and that of a fourth did not change at all.

The gain is not smaller next to hamlets, as the first channel alone would predict: schools with a hamlet or village within a kilometer gained 6.1 points (2.6), against 3.2 (1.3) for isolated schools, although the difference is imprecise ($p = 0.26$). These comparisons cannot separate light at the school from light at home. The cleanest test would compare isolated schools that received a solar panel, where the light stayed at the school (Figure 7), with isolated schools connected to the grid. Isolated schools gained 3.2 points (1.3) from the grid and 1.4 (3.9) from a solar panel; with only 80 isolated schools receiving a panel, the comparison cannot tell the two apart.

Night lights, which studies of electrification use as an outcome (Schmidt and Moradi, 2026; Figueiredo Walter and Moneke, 2024), cannot measure what happened at the school. Among

[40] The split is at the median share of children aged 12 to 14 who worked in the school's census zone in 1996. The difference, −4.4 (3.4), goes against the hypothesis; its bootstrap confidence interval rules out a difference larger than 3.2 points in its favor, though not a smaller one. In the censuses, connecting the zone's school did not detectably reduce child work in it either.

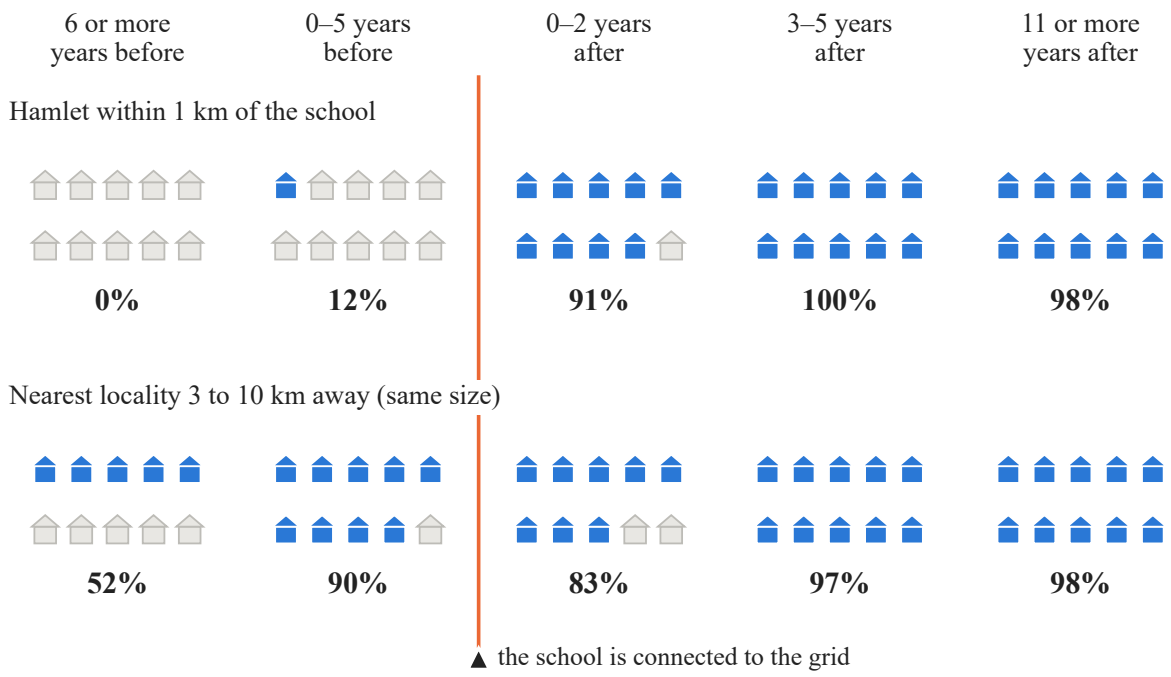


Figure 6: Grid electricity in the dwellings of hamlets next to rural schools, by years between the school's connection and the census.

Notes: Mean share of occupied dwellings with grid electricity in the nearest census locality; each icon is 10 percent, rounded. Hamlet: a locality within 1 km of the school with 5 to 100 occupied dwellings (roughly fewer than 300 people); comparison: the nearest locality of the same size, 3 to 10 km away. Censuses of 1996, 2011, and 2023; schools with a high-certainty connection date and, before the connection, only censuses in which the school was open. Schools: 37 in the hamlet row and 66 in the comparison row. Source: own elaboration based on ANEP and UTE records and INE's 1996, 2011, and 2023 censuses.

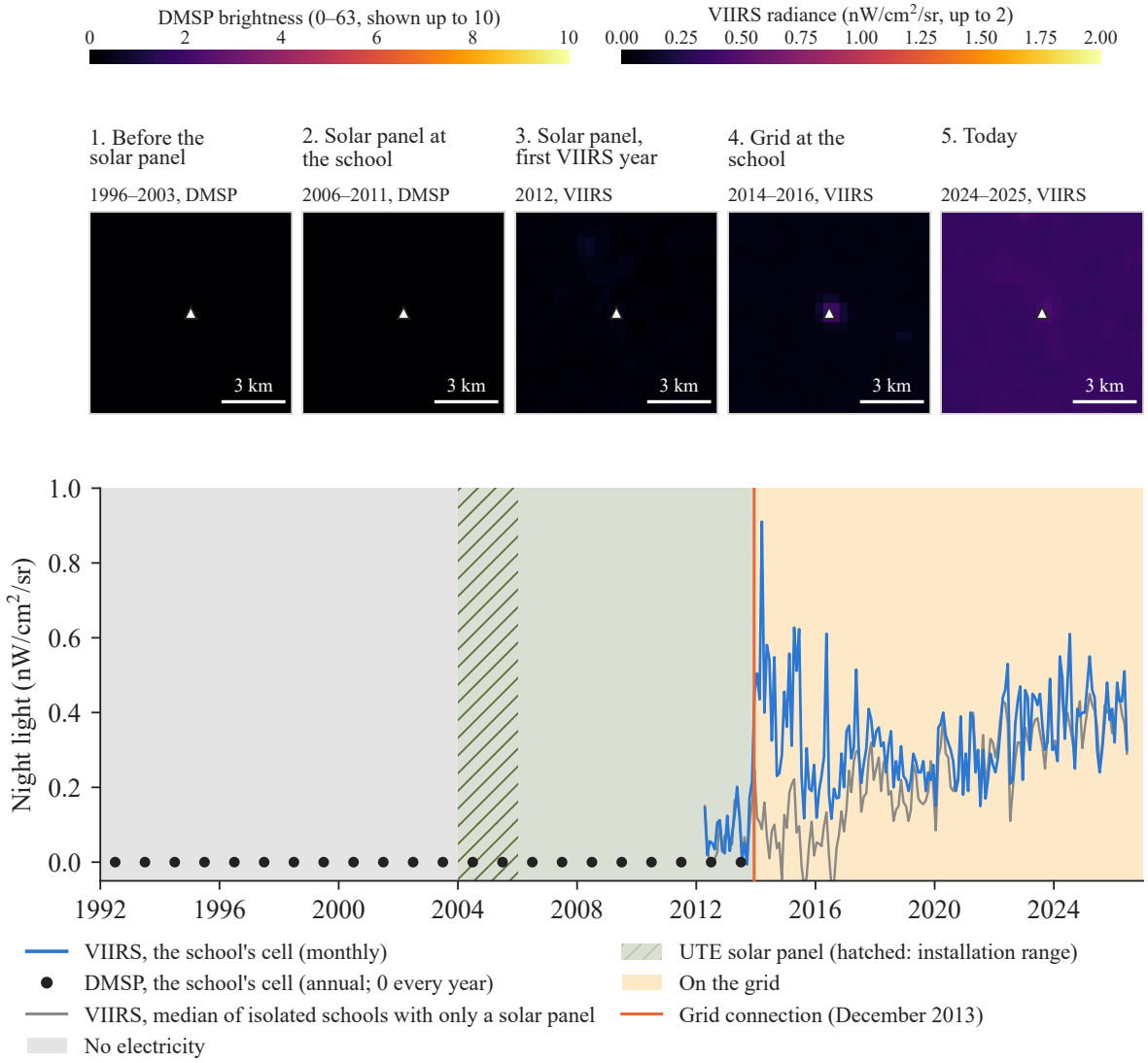


Figure 7: Night light at an isolated rural school with no electricity, then a solar panel, then the grid, 1992–2026.

Notes: A school in Rivera, 20 km from the nearest locality: UTE solar panel installed in 2004–2005 (hatched: the range of installation years), grid connection in December 2013. Top: mean night light around the school in each period, 10 km across (triangle: the school). Bottom: annual DMSP brightness of the school's satellite cell, 0 in every year from 1992 to 2013, and monthly VIIRS radiance, in nW/cm^2/sr, against the median of 66 isolated schools with only a solar panel. In 2017 the VIIRS product shifts up almost everywhere. The population around the school did not grow: in the rural part of its census section, residents fell from about 1,030 in the 2011 census to about 950 in 2023. About three dwellings, with some 10 people, stand within 1 km of the school, according to building footprints (Sirko et al., 2021) calibrated with the censuses, and the school's light in 2018–2020 is that of schools on the grid with as few people around. The school was chosen because VIIRS sees its connection: among the 38 isolated schools connected in 2013–2021, it has the largest rise. Source: own elaboration based on ANEP and UTE records and DMSP and VIIRS night lights (Elvidge et al., 1997, 2013).

the schools connected to the grid in 2012–2021, the light of the school's own satellite cell rose by only about 0.05 nW/cm^2/sr (standard error 0.03) in its first year with the grid, while the cell of a school next to a hamlet of 10 to 30 dwellings with the grid shines at about 1 nW/cm^2/sr, against 0.2 at isolated schools with only a solar panel. What the satellites see around a school is the settlement around it, not the school.[41]

6 Conclusion

Electricity got more of Uruguay's rural students to school regularly. Once a rural school was connected to the grid, the share of its students who reached satisfactory attendance rose by about 4 percentage points over its first two school years with electricity, from an average of 84 percent in school-years without electricity. Almost all of the gain came from students who would otherwise have attended irregularly, and it rests on the connections of the 1990s, years before Plan Ceibal's laptops reached rural classrooms.

The result is narrower than it may seem. Its sign is robust, but its size less so: some specifications give 2.4 to 3.0 points. Enrollment, the number of teachers, and school closure did not detectably change, and repetition shows no effect we can interpret. Solar panels produced no detectable gain, but their estimates are too imprecise to rule out effects as large as the grid's. Two further pieces of evidence point in the same direction, although neither is causal on its own: sixth graders in connected schools scored higher, if imprecisely, in the 1996 national assessment, and in census zones whose school was connected, the share of young people aged 18 to 24 who had completed lower secondary school rose from 18 to 40 percent, about 10 percentage points more than in zones whose school was still off the grid.

How electricity raised attendance remains open. The gain is larger, and lasts longer, far from larger localities, and it is not smaller next to hamlets, where the grid reached the dwellings at about the same time as the school, although neither difference is precisely estimated. Our comparisons cannot separate light at the school from light at home, and night lights cannot either: they register the settlements around schools, not the schools themselves.

These findings qualify the sobering reading of the literature. The studies that find no effect of electricity on schooling measure enrollment or test scores (Lee et al., 2020; Burlig and Preonas, 2024; Schmidt and Moradi, 2026), and the school-level evaluations measure examination results (Seo, 2017; Berkouwer et al., 2018; Koima, 2024); none measures how often students attend. In Uruguay, enrollment did not respond either, with little room to grow when nearly all children of primary-school age already attended school, but attendance did. The gain is about half the 8.4 points that solar lanterns produced at surprise school visits in rural Bangladesh (Kudo et al., 2019), from a much higher base.

The census result matters for what it suggests about mobility across generations. In 1996, 72 percent of rural children of primary-school age had a mother with at most primary education, and only 6 percent one who had completed upper secondary school; in ANEP's records for the schools of our sample, 61 percent of students still had a mother with at most primary education in 2010.[42] More young people completing lower secondary school means more children reaching a level of schooling that most of their mothers had not reached. It is also the hardest margin to move in rural Uruguay. Lower-secondary schools are far fewer than primary schools, so continuing to study means traveling (Section 2.1), and by lower-secondary age many children

[41] The satellite passes after midnight, so it sees outdoor or security lights rather than a school in use.

[42] Own calculations from the 1996 census microdata (INE, 1996), for children aged 6 to 11 in the dispersed rural area (Section 2.1). The 2010 share weights the schools of the analysis sample by their enrollment in grades 1 to 6.

are expected to work or help at home (Trujillo, 2008). Qualitative evidence suggests that, although students and their families value schooling, its perceived instrumental value may be limited: rural lower-secondary students report that their studies become secondary when the household has a secure source of work, and that the school curriculum bears little relation to the tasks they do at home (Trujillo, 2008).

Three implications for policy follow, within the limits of this evidence. First, electrifying the school itself can raise attendance even where there is no hamlet around it for the grid to reach: the gain is present in isolated schools, with no hamlet or village within a kilometer. Second, the gain came before laptops reached rural classrooms, so the value of electrifying a school need not depend on the devices that come with it. Third, our estimates cannot tell whether a solar panel delivers what the grid does; where the grid does not reach, that remains the open question for programs such as Lights to Learn.

The evidence has limits. We measure attendance, not learning: the national assessment reached too few schools without electricity to estimate an effect on test scores. The grid did not arrive at random, so our estimates rest on connected and comparison schools following parallel paths, which the placebo estimates support but cannot prove. The census results compare zones, not students, with a single census before the connections and with people who may have moved. Three questions remain open: whether solar panels deliver the gains of the grid, which requires more schools than the panels reached in Uruguay; whether more regular attendance translates into learning, which requires following students into national assessments; and whether the gain comes from light at the school or at home, which requires knowing when the households around each school were connected.

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A Additional descriptive evidence

This appendix collects the descriptive evidence behind Section 2. Appendix Table 1 describes the schools of the estimation sample in 1992–1994, and Appendix Table 4 defines each of its rows. Appendix Tables 2 and 3 list every source with its period and coverage, and the text that follows details their contents. Appendix Figure 8 shows the source of electricity of the schools by distance to the nearest populated place, and Appendix Figure 9 follows, year by year, each school that ever had a solar panel. Appendix Figure 10 shows the schools connected to the grid each year, by their previous source of electricity.

	Median	Mean	P10	P90	Schools
Panel A. Educational outcomes, 1992–1994					
Students, grades 1–6	15.3	20.6	6.3	41.7	1,095
Attendance, 141 days or more (%)	89.6	84.1	61.1	100.0	1,095
Attendance, 71 to 140 days (%)	8.7	13.6	0.0	33.4	1,095
Attendance, 1 to 70 days (%)	0.0	2.3	0.0	6.1	1,095
Repetition (%)	9.9	11.1	1.8	21.2	1,095
Panel B. The school, 1992–1994					
Teachers, grades 1–6	1.0	1.5	1.0	2.3	1,095
One-teacher school (%)	–	59.9	–	–	1,095
Panel C. Socioeconomic status					
Mother with primary education or less (%)	64.3	62.8	33.3	100.0	1,072
Unmet basic needs index (0–100)	24.0	27.0	6.3	53.6	1,051
Panel D. Electricity in 1992					
Connected to the grid in 1992 (%)	–	43.1	–	–	1,091
Solar panel, no grid, in 1992 (%)	–	5.1	–	–	1,091
No electricity in 1992 (%)	–	51.8	–	–	1,091
Panel E. Isolation					
Km to the nearest INE locality, any size	6.1	7.4	0.0	16.2	1,102
Km to the nearest INE locality, 5,000+ inhabitants	23.0	28.6	5.3	60.6	1,102
Isolated school (%)	–	78.8	–	–	1,102
Very low density rural GHSL cell (%)	–	85.8	–	–	1,102
Panel F. People and buildings around the school					
People within 1 km (GHSL)	15	64	1	176	1,102
Buildings within 1 km (Open Buildings)	33	97	5	219	1,063

Table 1: Rural schools: descriptive statistics.

Notes: One observation per school; broad rural sample (1,102 schools). Columns report the median, mean, 10th and 90th percentiles across schools, and the number of schools with data; in rows with only a mean, the mean is the share of schools, and the other percentages are shares of students in each school. Panels A and B average each school's open years in 1992–1994. In 1992, the share with a solar panel is an upper bound and the share with no electricity a lower bound, because UTE dates early panels only by range. Distances are to the nearest INE locality of each size in the 2011 census; an isolated school has none within 1 km. GHSL refers to 1990 and Open Buildings to about 2020. Household indicators come from ANEP surveys of 2005 and 2010, after most connections. Appendix Table 4 defines each row. Source: own elaboration based on ANEP and UTE records, INE's 2011 census, GHSL, and Google Open Buildings.

Tables 2 and 3 list the sources of the panel, one row per source. ANEP's household surveys of 2005, 2010, 2015, and 2020 come with ANEP's school records. In UTE's supplies, supplies connected before 1 May 1992 generally carry that date, and the dates that UTE re-verified in 2024 take precedence over the first service. The Department of Rural Education provided a list of the Lights to Learn schools (created in 2014 and sent in 2022), a list of the schools with panels in March 2022, and a note. The documents on single schools are UTE's 2019 departmental reports, UTE's list of the schools of the 2009 agreement between ANEP and UTE, minutes of departmental legislatures, and UTE news releases; the counts in that row are the schools with a lower or an upper bound from any document of Panel A. For Plan Ceibal, the yearbook takes its table from Plan Ceibal. The SINAE regions are Centro, Este, Litoral Norte, Litoral Sur, Metropolitana, and Noreste. The police section comes from a layer that joined the schools' points to police sections (1,172 schools of the panel; the origin of that layer is not established) or, for 165 schools, from the code most common among the other schools in the same polygon of INE's 2010 map. UTE's records were delivered under the access to public information law (Law 18.381).

Panel C, in Table 3, links data to each school through its coordinates: own, for 1,179 schools, or from secondary sources, for 173 (departmental maps of 2012, online map listings, OpenStreetMap, and the hamlet each school reports). The 1996 localities are drawn with INE's 2004 maps and its correspondence between the census zones of 1996 and 2004. From GHSL we use GHS-POP and GHS-BUILT-S at 3 arc-seconds and the settlement class (GHS-SMOD) of the 1 km cell, every five years. From Open Buildings we use the footprints with confidence 0.70 or more (release of May 2023; imagery dates not reported) and the yearly presence of buildings (Temporal V1); within 1 km, the counts cover only the schools of the analysis sample with their own coordinates. From VIIRS we use the monthly composites corrected for stray light, averaged by year. NASA POWER is based on MERRA-2. The cadastre is the current one, not that of the 1990s. VIIRS, DMSP, Open Buildings, and GHS-SMOD were processed in Google Earth Engine (Gorelick et al., 2017). Besides the sources cited in each row, the two tables draw on INE (2004, 2008), Pesaresi and Politis (2023); Schiavina et al. (2023a), Sirko et al. (2023), Baugh et al. (2010), Mills et al. (2013), Gelaro et al. (2017), Harrigan et al. (2020), notocactus.eu (2012), OpenStreetMap contributors (2026), and GeoNames (2026), accessed between September and October 2026.

Source	What it provides	Unit or resolution	Period	Schools (panel / sample)	Linked by
Panel A. School records and electricity dates					
ANEP–DIEE, school records (ANEP–DIEE, 2021)	Enrollment, students by days attended, repetition, teachers; surveys of students' households, 2005, 2010, 2015, and 2020	School-year	1992–2021; teachers to 2015	1,380 / 1,159	School code (RUEE)
ANEP, assessment of sixth graders (ANEP, 1996)	Electricity reported by principals and households	Student	1996	219 / 200	School code (RUEE)
UTE, supplies of ANEP's schools (UTE, 2024c)	First service at the school's premises; dates re-verified for 128 schools	Supply	First services of 1991–2022	1,044 / 1,026	Department and school number
UTE, solar panels (UTE, 2024a)	Year or range of years of installation	School	1992–2010	272 / 272	Department and school number
UTE, Lights to Learn (solar panels) (UTE, 2022)	Schools of the program; date of the panel; year of a later grid connection	School	2013–2022	85 / 85	Department and school number
ANEP, Department of Rural Education (ANEP–DEMR, 2022)	Lights to Learn schools; panels still in use in 2022; electricity in every school from 2015	School	2013–2022	73 / 73	School code (RUEE)
Documents on single schools (UTE, 2019, 2024b)	Lower or upper bound on the connection date	School	Bounds dated 1992–2026	Lower 290 / 289; upper 223 / 222	Department and school number
Panel B. Plan Ceibal and the groupings of schools					
MEC, Plan Ceibal (MEC, 2009)	Month in which the laptops reached each department	Department	2007–2009	1,380 / 1,159	Department
SINAE, regions (SINAE, n.d.)	Region of the school's department	Six groups of departments	Undated	1,380 / 1,159	Department
Police sections (INE, 2010)	Police section of the school, the unit of the clusters	Polygon	2010	1,337 / 1,156	Coordinates

Table 2: Data sources.

Notes: One row per source. Schools: the number of the 1,380 schools of the panel and of the 1,159 schools of the analysis sample with data from the source. Linked by: ANEP's school code (RUEE); the department and the school's number, for the sources without the code; the department; or the school's coordinates (Panel C, in Table 3). The text of this appendix details the contents of each source. Source: own elaboration based on the sources cited in each row.

Source	What it provides	Unit or resolution	Period	Schools (panel / sample)	Linked by
Panel C. Data linked through the school's location					
ANEP, school layer (ANEP, 2024); secondary sources	Coordinates of the school	Point	Layer of 2024; secondary sources of 2012–2026	Own 1,179 / 1,119; secondary 173 / 40	—
INE, 1996 census (INE, 1996)	Localities and their population; rural residents and dwellings	Locality; census area	1996	1,352 / 1,159; areas 944 / 907	Coordinates
INE, 2011 census (INE, 2011a,b)	Localities and their population; rural residents and dwellings	Locality; census segment	2011	1,352 / 1,159; segments 962 / 924	Coordinates
INE, 2023 census (INE, 2026)	Rural residents and dwellings	Census segment (2011 code)	2023	750 / 723	Coordinates
JRC, GHSL R2023A (Schiavina et al., 2023b)	Population and built-up area within 1, 5, and 10 km; settlement class	About 100 m; 1 km (class)	Epochs 1975–2020; built-up area from 1990	1,352 / 1,159	Coordinates
Google, Open Buildings (Sirko et al., 2021)	Buildings within 1 and 5 km, and by year	Footprint	Release of 2023; 2016–2023	1,352 / 1,159; 1 km 1,119 / 1,119	Coordinates
EOG, DMSP-OLS stable lights (Elvidge et al., 1997)	Night lights at and around the school	About 1 km	1992–2013	1,352 / 1,159	Coordinates
EOG, VIIRS monthly composites (Elvidge et al., 2013)	Night radiance at and around the school	About 460 m	2014–2023	1,352 / 1,159	Coordinates
CHG, CHIRPS v2.0 (Funk et al., 2015)	Daily rainfall	0.05°	1992–2021	1,352 / 1,159	Coordinates
NASA, POWER (NASA Langley Research Center, 2026)	Daily maximum and minimum temperature	0.5° by 0.625°	1992–2021	1,352 / 1,159	Coordinates
ECMWF, GloFAS v4.0 (Zsoter, 2019)	Daily discharge of the largest river within 20 km	0.05°	1992–2021	1,352 / 1,159	Coordinates
DNC, rural cadastre (DNC, 2026)	Area of the rural parcels within 5 km	Parcel	Current (queried in 2026)	1,217 / 1,157	Coordinates
MIEM, hydroelectric dams (MIEM, 2026)	Distance to the nearest of the four dams	Point	Built 1945–1982	1,339 / 1,157	Coordinates

Table 3: Data sources, continued.

Notes: One row per source. Schools: the number of the 1,380 schools of the panel and of the 1,159 schools of the analysis sample with data from the source. For the 1996 and 2011 censuses, the first count is the schools with a distance to the census localities and the second the schools placed in a dispersed rural census area; in 2023, the schools whose 2011 segment code has residents. The 28 schools located only by their department have no coordinate-based data. Image and data processing of DMSP and VIIRS by the Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines; DMSP data collected by the US Air Force Weather Agency. OpenStreetMap data © OpenStreetMap contributors. Source: own elaboration based on the sources cited in each row and in the text of this appendix.

Row	Definition
Panel A. Educational outcomes, 1992–1994	
Students, grades 1–6	Final enrollment in grades 1–6, mean of each school over its open years in 1992–1994 (in 6 school-years, the sum of the students enrolled by group).
Attendance	Share of students of grades 1–6 attending 141 days or more, 71 to 140 days, and 1 to 70 days of the school year.
Repetition	Share of students of grades 1–6 who repeated the grade.
Panel B. The school, 1992–1994	
Teachers, grades 1–6	Teachers of grades 1–6, mean of each school over its open years in 1992–1994.
One-teacher school	A single teacher of grades 1–6 in at least half of the school's open years in 1992–1994.
Panel C. Socioeconomic status	
Mother with primary education or less	Share of students whose mother completed primary school or less (ANEP survey of 2010).
Unmet basic needs index	Of the students' households: 0 = no student with an unmet need, 25 = all with one, 100 = all with four (ANEP survey of 2005).
Panel D. Electricity in 1992	
Electricity in 1992	Source of electricity in 1992, for the 1,091 schools open that year whose source is known. 452 of the 470 schools on the grid in 1992 carry UTE's filler date (connected by 1 May 1992, exact date unknown). Solar panel: a solar panel installed by UTE and no grid; UTE dates the 56 panels of 1992 only by the range 1992–1995, and we count them from 1992.
Panel E. Isolation	
Km to the nearest INE locality	Distance from the school to the boundary of the nearest locality of each size in the 2011 census (0 if inside). With the 1996 census, the median distance to a locality of 5,000 inhabitants or more is 24.4 km in this sample and 23.4 km in the 1,159 schools of the analysis sample, where the estimates split the schools at that median.
Isolated school	No INE locality of the 2011 census within 1 km; with the 1996 census, 86.4% of the schools are isolated.
Very low density rural GHSL cell	The degree of urbanization of the school's 1 km cell is class 11 (very low density rural), 1990 epoch.
Panel F. People and buildings around the school	
People within 1 km	Modeled residents, GHSL, 1990 epoch.
Buildings within 1 km	Open Buildings v3 footprints with confidence 0.70 or more, from imagery of about 2020 (2020–2023), excluding those within 50 m of the school (the school itself), for the schools with their own coordinates.

Table 4: Definitions of the rows of Appendix Table 1.

Notes: Both household surveys postdate most connections: 79% of the schools of the sample with no electricity in 1992 that got it later had it by 2004, and in 2005, 93% of the schools open that year had grid electricity or a solar panel. Between the 1996 and 2023 censuses, the number of dwellings in the school zones barely changed (median change −2%), while the median zone or census section lost 40–50% of its people. Reduced rural sample (990 schools, without the 15% least rural), medians (shares of schools for the indicators): Panel A, students 13.7; 141+ days 89.3%; 71–140 days 9.0%; 1–70 days 0.0%; repetition 9.6%; Panel B, teachers 1.0; one-teacher 65.9% of schools; Panel C, mother with primary or less 66.7%; unmet basic needs 25.0; Panel D, grid 38.0% of schools; solar panel 5.6% of schools; no electricity 56.4% of schools; Panel E, any size 7.0 km; 5,000+ 25.1 km; isolated 82.0% of schools; very low density rural 88.8% of schools; Panel F, people 12; buildings 28. Source: own elaboration based on ANEP and UTE records, INE's 1996, 2011, and 2023 censuses, GHSL, and Google Open Buildings.

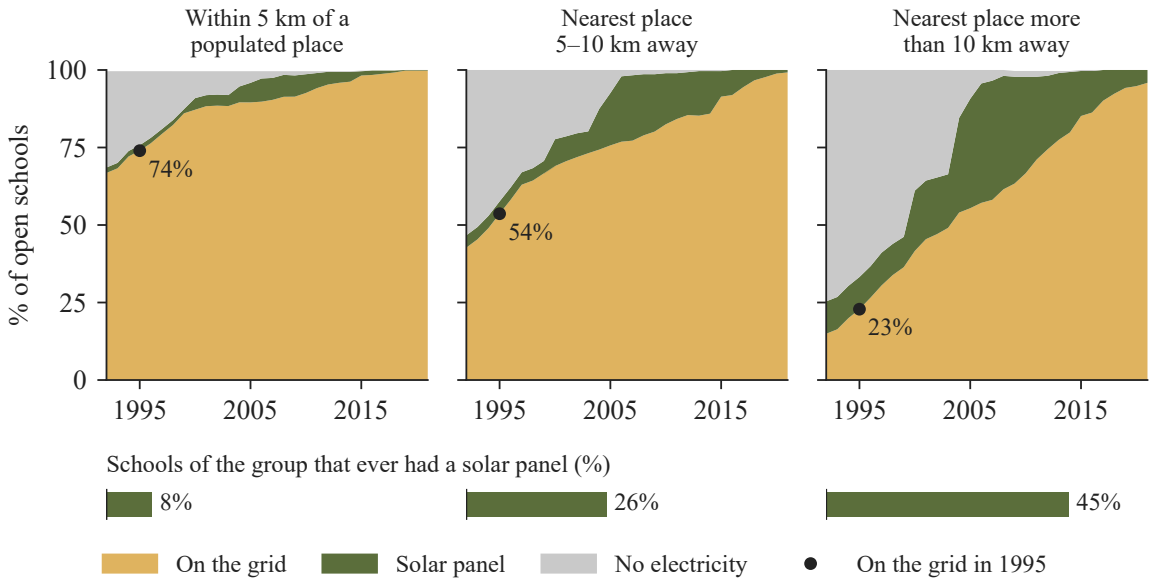


Figure 8: Source of electricity of rural schools by distance to the nearest populated place, 1992–2021.

Notes: Percentage of each group's open schools by source of electricity, dated as in Figure 1; solar panels are UTE's and Lights to Learn's. Dots: percentage on the grid in 1995. Bars: percentage of the group's schools that ever had a solar panel. Groups, by the distance to the nearest populated place (any locality delimited by the INE in the 2011 census, from a hamlet of a few dwellings to a city): within 5 km (519 schools), 5–10 km (302), and more than 10 km (338). Source: own elaboration based on ANEP and UTE records and INE's 2011 census.

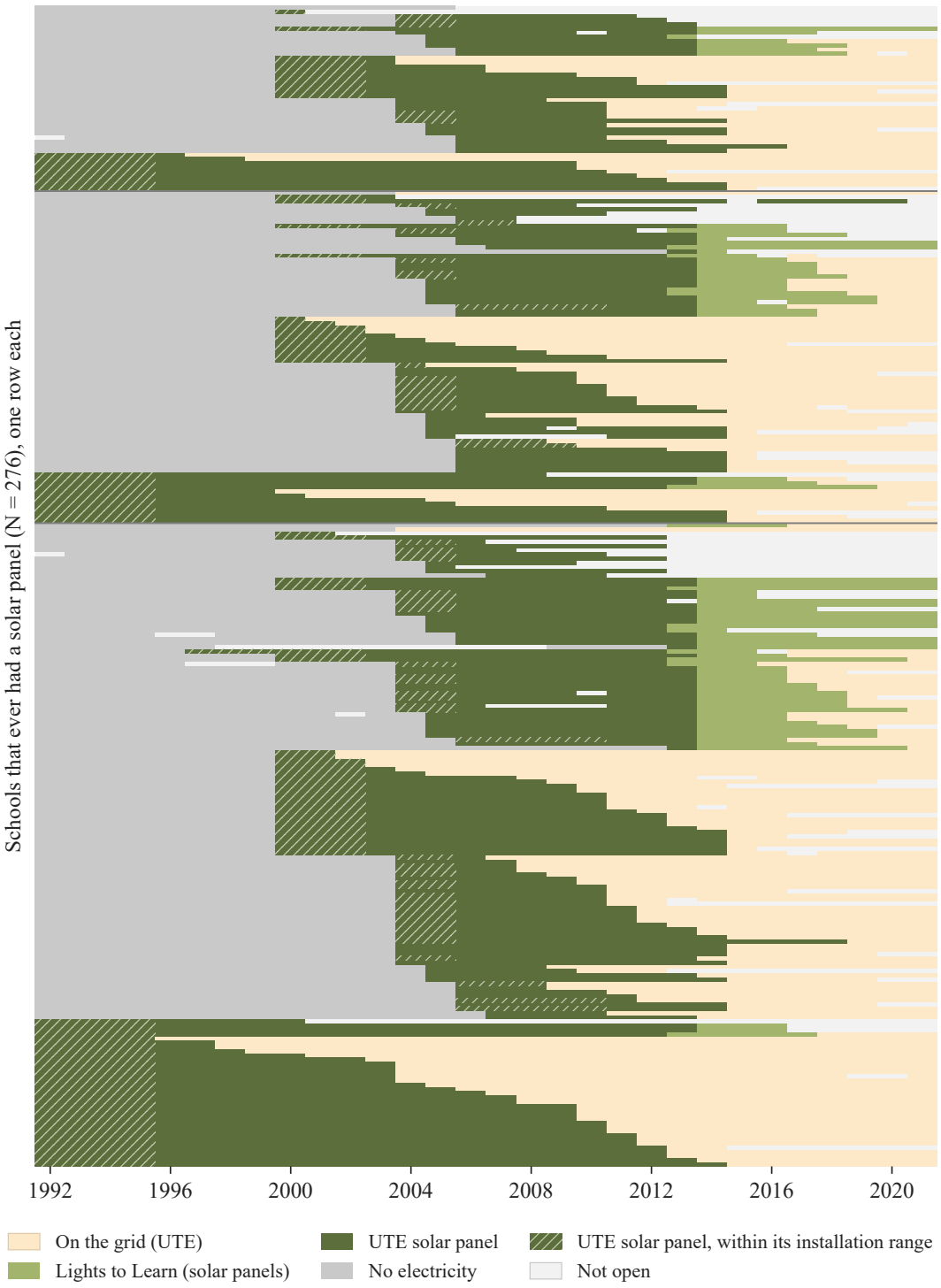


Figure 9: Source of electricity of each rural school that ever had a solar panel, by distance to the nearest locality.

Notes: One row per rural school that ever had a UTE or Lights to Learn solar panel (N = 276), grouped by distance to the nearest 2011-census locality and ordered by trajectory within each group. Groups, from top to bottom, separated by thin lines: within 5 km of a locality (44), 5–10 km (79), and more than 10 km (153). Source: own elaboration based on ANEP and UTE records and INE's 2011 census.

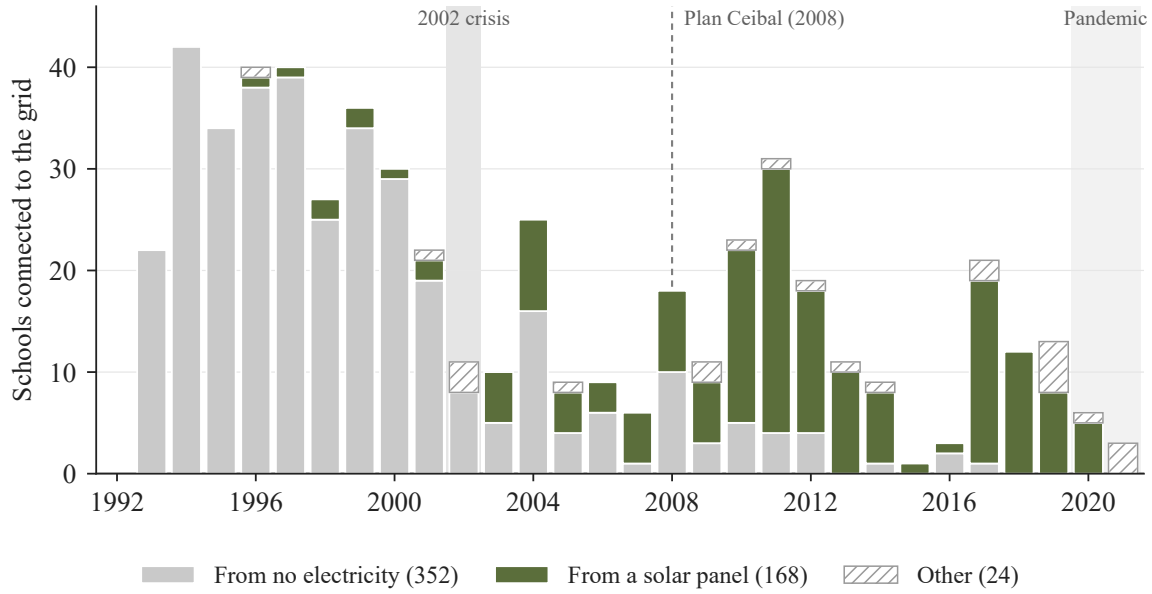


Figure 10: Schools connected to the grid each year, by their previous source of electricity (if any).

Notes: N = 1,159 schools of the analysis sample; 544 grid connections, 1993–2021. Other: connections not counted as a change of source in the estimates. The shaded year 2002 marks Uruguay's economic crisis; the dashed line, the arrival of Plan Ceibal; the shaded 2020–2021, the pandemic. Source: own elaboration based on ANEP and UTE records and MEC (Anuario Estadístico 2008).

B The rollout of the grid and the instruments

This appendix collects the evidence on how the grid reached rural schools, behind Section 3. Appendix Table 5 reports what predicted when a school was connected, and Appendix Figure 11 and Appendix Table 6 report the exploratory tests of two candidate instruments for the timing of connection.

Determinants of the grid rollout

Table 5 reports the discrete-time hazard of connection to the grid of the rural schools of the analysis sample without grid at the start of 1993, with predictors measured before 1993.

The results indicate that electrification timing was systematically related to pre-existing infrastructure, surrounding population, demographic growth, and initial school enrollment. Although selection on observable characteristics does not itself invalidate the difference-in-differences design, it raises concerns about parallel trends if these characteristics also predict differential changes in educational outcomes. In particular, differences in baseline enrollment and local demographic dynamics could generate divergent educational trajectories even in the absence of electrification.

In Table 5, one more school of the risk set, located only at the nearest INE locality, has no distance to the dams and is left out. The risk set includes only years in which the school appears in the records of ANEP (the public education administration), and leaving them is censoring. Standard deviations are those among the 586 schools, and a negative coefficient on a distance means that schools farther away connected later. The predictors are the distance to the nearest school already on the grid in 1992; the distance to the nearest lit cell in the 1992 DMSP night lights; the distance to the nearest hydro dam; the distance to the nearest INE locality (1996 census) of any size, of 5,000 inhabitants or more, and of 15,000 or more; the population within 5 km in 1990 (GHSL) and its growth since 1975; and enrollment in grades 1–6 in the first open year (1992 except for 7 schools that open in 1993–2004). Every column controls for the distance to the nearest other school (density of schools). Block tests are Wald F tests with the cluster-robust covariance, $F(k, G-1)$, conditional on the other blocks; the 1992 night lights also mark lit localities, and testing the distances to localities together with the distance to a lit cell gives $F = 3.34$ [0.012], 3.61 [0.007], 1.08 [0.373], and 3.98 [0.004] in columns (1), (2a), (2b), and (3). The school block, enrollment alone, gives $F = 19.25$ [0.000], 22.45 [0.000], 1.84 [0.179], and 19.61 [0.000], the square of the t statistic of its coefficient. The row (2a) = (2b) tests that the nine predictors are the same in both columns, in one model with every regressor interacted with the panel state and region × year × state fixed effects. In column (1) the coefficient on the distance to a school on the grid in 1992 depends on the density control (without it, −0.900 (0.542)), so it is read with its block. School-years alone in their fixed-effect cell are dropped (12, 19, 25 and 0 in columns (1), (2a), (2b) and (3)); the counts are those of the estimated sample. For 23 schools without an original coordinate, the location comes from secondary sources. The coefficients describe which schools were connected first.

Exploratory instrumental-variable tests

Given the non-random timing of rural electrification documented in Table 1, we explored whether geographical characteristics could provide an alternative source of identifying variation. Specifically, we investigated two candidate instruments intended to explain differences in the timing of grid connection across schools. For these instruments to identify the causal effects

	Region × year (1)	No electricity (2a)	With a solar panel (2b)	Year and department (3)
Existing grid				
Distance to a school on the grid in 1992 (log km)	-1.169**	-1.161**	-0.986	-0.802*
	(0.501)	(0.538)	(0.920)	(0.483)
Distance to a lit cell, DMSP 1992 (log km)	-1.491**	-1.530**	-0.556	-2.073***
	(0.672)	(0.712)	(1.451)	(0.674)
Distance to the nearest hydro dam (log km)	-0.507	-1.204	1.978*	-1.775*
	(0.778)	(0.831)	(1.060)	(1.009)
Distance to localities				
Distance to the nearest 1996 locality (log km)	0.565	0.388	1.330*	0.424
	(0.435)	(0.575)	(0.763)	(0.468)
Distance to a locality of 5,000+, 1996 (log km)	-0.847	-0.845	-1.590	-0.273
	(0.764)	(0.870)	(1.596)	(0.653)
Distance to a locality of 15,000+, 1996 (log km)	0.277	0.131	1.212	0.115
	(0.688)	(0.809)	(1.253)	(0.643)
Demand				
Population within 5 km, GHSL 1990 (log)	2.753***	1.638***	4.557***	2.760***
	(0.529)	(0.624)	(0.906)	(0.519)
Population growth within 5 km, 1975-1990 (log)	0.790**	0.796*	0.081	0.834**
	(0.380)	(0.432)	(0.559)	(0.407)
School				
Enrollment, first open year (log)	1.855***	2.370***	0.997	1.883***
	(0.423)	(0.500)	(0.735)	(0.425)
Control				
Distance to the nearest other school (log km): density	0.835**	-0.119	1.430***	0.556
	(0.361)	(0.559)	(0.503)	(0.353)
Joint tests: F [p]				
Existing grid	4.16 [0.007]	4.21 [0.007]	1.99 [0.122]	6.26 [0.000]
Distance to localities	0.92 [0.433]	0.63 [0.595]	1.13 [0.340]	0.32 [0.814]
Demand	16.17 [0.000]	5.35 [0.006]	12.69 [0.000]	16.23 [0.000]
All nine predictors	13.17 [0.000]	9.68 [0.000]	5.45 [0.000]	14.28 [0.000]
(2a) = (2b), the nine predictors		2.22 [0.023]		
Region × year fixed effects	Yes	Yes	Yes	No
Year and department fixed effects	No	No	No	Yes
Mean of the outcome (connections per 100 school-years)	7.86	7.99	7.48	7.87
Schools	586	536	225	586
Connections	532	345	181	534
Observations (school-years)	6,772	4,319	2,421	6,784
Clusters (police sections)	174	170	89	174

Table 5: Predictors of the timing of grid connection of rural schools, 1993–2021.

Notes: Linear probability model of the discrete-time hazard of connection to the grid. Unit: school-year of the risk set, the 586 rural schools of the analysis sample without grid at the start of 1993, until the year they connect or, if they never connect, until their last documented year without grid or 2021. The outcome is an indicator for connection in that year, times 100, so coefficients are percentage points of the annual probability of connection per standard deviation of the predictor. Predictors are measured before 1993 or are fixed geography, and enter as $\log(1 + x)$. Columns (2a) and (2b) split the school-years by whether the school had a solar panel before the year. Standard errors clustered on police section in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The coefficients are not causal effects. Source: own elaboration based on ANEP and UTE records, GHSL, DMSP night lights, INE's censuses, and MIEM.

of electrification, they must predict grid access (relevance) and affect educational outcomes exclusively through electrification (exclusion restriction). This exploratory exercise evaluates the feasibility of an alternative identification strategy rather than providing an independent validation of the assumptions underlying our main difference-in-differences design.

The hydroelectric dams. Uruguay's four hydroelectric dams (Rincón del Bonete, 1945; Baygorria, 1960; Salto Grande, 1979; and Palmar, 1982) were built where hydrology dictated, and the lines that carried their power could have brought the grid earlier to the rural areas around them. The instrument is the distance from each school to the nearest dam (Figure 11). There are two reasons to doubt it from the start. The dams deliver power at 150 and 500 kV, and rural distribution starts at substations in larger localities. And the works moved population, which may affect schooling directly: Paso de los Toros grew with the construction of Rincón del Bonete, and the old locality of Constitución, in the department of Salto, and part of Belén were relocated for the Salto Grande reservoir. For this reason we also repeat the test without the schools within 100 km of a dam, the largest circles in Figure 11. The radius comes from a preliminary analysis, in which schools up to about 100 km from a dam had a better sociocultural context and better-equipped households (ANEP's context surveys) than schools farther away.

The size of rural parcels. Rural extensions of the grid were often built at the expense of those who requested them, who hired a contractor registered with UTE. Large landowners could have advanced the connection by paying for it, or the scarcity of neighbors could have delayed it. The instrument is the area of the largest rural parcel that touches the 5 km disc around the school, from the current rural cadastre of the National Directorate of Cadastre (DNC). This is a weaker candidate for several reasons. First, the cadastre describes contemporary parcel boundaries rather than those prevailing in 1993. Changes in landholding patterns over the study period may introduce measurement error, and changes associated with electrification or local economic development could further undermine the instrument's validity. Second, the cadastre does not identify owners, meaning that individual parcels cannot necessarily be interpreted as agricultural holdings. Finally, larger parcels may be associated with lower population density and fewer families, potentially affecting school enrollment independently of electrification.

Specification. The sample is the risk set of Table 5: the 586 rural schools of the analysis sample without grid at the start of 1993, in every year they were open from 1993 to 2021 (16,119 school-years; a school with no grid date leaves the sample after its last year documented without grid). Each instrument Z_i is the standardized $\log(1+x)$ of the school's distance (in km) or area (in hectares). Because it is fixed for each school, the school fixed effect absorbs its level, so it enters interacted with time:

$$D_{it} = \pi\, Z_i \tfrac{t-1993}{10} + \alpha_i + \lambda_{r(i)t} + v_{it},$$
$$Y_{it} = \delta\, Z_i \tfrac{t-1993}{10} + \alpha_i + \lambda_{r(i)t} + e_{it},$$

where D_{it} indicates that school i is on the grid in year t, Y_{it} is attendance (141 days or more), repetition or enrollment, α_i are school fixed effects and $\lambda_{r(i)t}$ are region-by-year fixed effects, the comparison of the main design. Standard errors are clustered by police section (174 clusters). Because the instruments are time-invariant, school fixed effects absorb their levels. Identification therefore relies on whether schools with different geographical characteristics experienced systematically different changes in grid access over time, conditional on region-by-year fixed effects. The exclusion restriction requires that the instrument-related differences in educational trends operate exclusively through electrification. This assumption may be violated if proximity to hydroelectric dams or rural parcel size is associated with differential demographic or

economic trends that independently affect educational outcomes. Relevance is measured with the effective F statistic of Montiel Olea and Pflueger (2013), which with one instrument equals the robust first-stage F; below 23.1 the instrument is weak. With a weak instrument the usual two-stage least squares t-test is not valid (Lee et al., 2022), so the test of an effect of the grid is the Anderson and Rubin (1949) test, which with one instrument is the test of the reduced form ($\delta = 0$), together with its 95% confidence set.

Results. Table 6 reports both tests. Both instruments are weak: within region, the effective F is at most 7.6 for the dams and 2.4 for the parcels across the three outcomes (5.5 and 4.2 with year instead of region-by-year fixed effects), far below 23.1. The reduced forms also show signs that the exclusion restriction fails. With the dams, the reduced form for repetition rejects zero (p = 0.001), despite the weak first stage. The Anderson–Rubin 95% confidence set implies an increase in repetition of at least 6.7 percentage points, compared with a sample mean of 5.8%. This lower bound exceeds the mean repetition rate, raising substantial concerns about the plausibility of the implied effect. However, this result does not independently establish a violation of the exclusion restriction. Leaving out the 165 schools within 100 km of a dam (Panel C) does not strengthen the instrument (effective F at most 4.4), and the reduced form for repetition still rejects zero ($p = 0.010$), again only toward implausibly large effects (an increase of 6.5 percentage points or more). The same holds without the schools within any radius from 10 to 100 km, every 10 km: the effective F is at most 5.9, and repetition rejects zero at every radius ($p \leq 0.010$). The sign of a failed exclusion restriction does not come only from the schools closest to the dams. With the parcels, the reduced form for enrollment is negative and significant ($p = 0.004$): relative to the other schools of the same region, enrollment fell over time around the schools surrounded by large parcels, which is consistent with a direct channel through the number of families. The corresponding Anderson–Rubin confidence set includes only effects below -1.40 or above 3.27 log points, indicating that the instrument provides no economically informative estimate of the effect of electrification on enrollment. The reduced form for repetition with the parcels also rejects zero ($p = 0.016$), again only toward very large effects. For attendance, and for enrollment with the dams, the sets contain zero and are too wide to be informative. Neither candidate instrument provides sufficiently strong and credible variation to identify the effects of electrification reliably. Weak first stages, combined with concerns about potential direct geographical and demographic channels, prevent us from pursuing either instrumental-variable strategy. We therefore retain the difference-in-differences design as our main empirical approach. Its credibility, however, rests on the plausibility of parallel trends and must be assessed independently of these unsuccessful instrumental-variable exercises.

Definitions in Table 6. The sample of Panels A and B is the risk set of the specification; a school with no grid date leaves after its last year documented without grid. Panel B uses the hectares of the largest rural parcel touching the 5 km disc around the school, from the current rural cadastre (DNC). In the first stage, the indicator of being on the grid is dated for each outcome as in the main design; with school fixed effects its sign does not say whether connection came earlier or later. Attendance is the percentage of students in grades 1–6 attending 141 days or more; repetition, the percentage of students in grades 1–6 who repeat; enrollment, $\log(1 + \text{students in grades 1–6})$. The effective F (Montiel Olea and Pflueger, 2013) equals the robust first-stage F with one instrument; its critical value for a worst-case bias of 10% is 23.1. The Anderson–Rubin test of no effect of the grid (Anderson and Rubin, 1949) is the test of the reduced form. With year instead of region $\times$ year fixed effects, the effective F is 5.5, 5.4, 2.6 (Panel A) and 4.2, 4.2, 1.7 (Panel B). In Panel C, the instrument is standardized again among the 421 schools that remain.

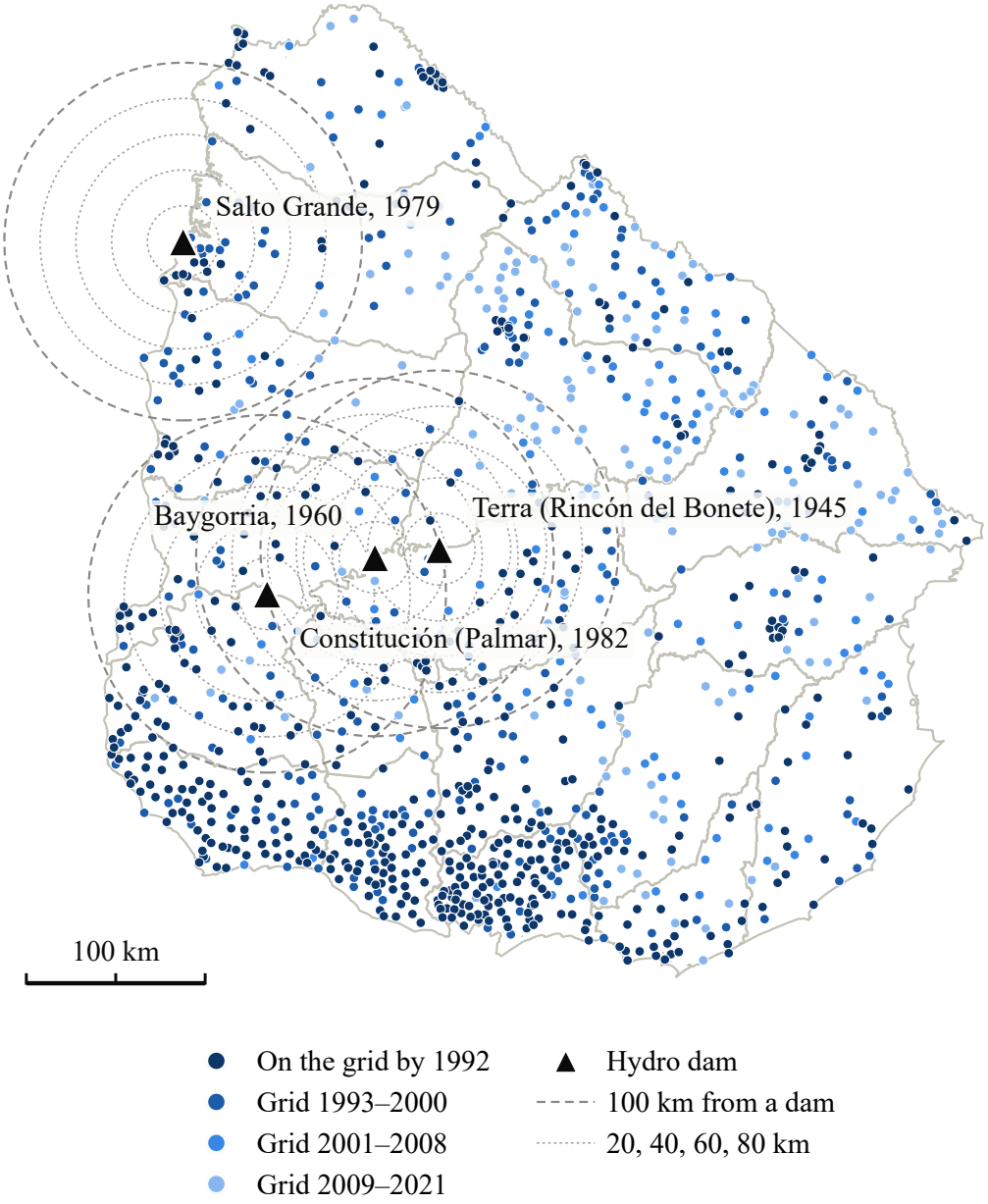


Figure 11: Year of grid connection of rural schools and location of the hydroelectric dams.

Notes: The 1,065 rural schools of the analysis sample connected to the grid by 2021 (not shown: the 94 with no grid by 2021 or no grid date); for 34, an estimated point. Dashed circles: 20, 40, 60, 80 and 100 km from each dam. Panel C of Table 6 leaves out the schools within 100 km; the text reports the test for every radius from 10 to 100 km. Source: own elaboration based on ANEP and UTE records, INE (department outlines), and MIEM (dams).

	Attendance (141 days or more, pp) (1)	Repetition (pp) (2)	Enrollment (log) (3)
Panel A. Distance to the nearest hydro dam			
First stage: instrument	0.027***	0.027***	0.025***
	(0.010)	(0.010)	(0.009)
Effective F (critical value 23.1)	7.60	7.62	7.25
Reduced form: instrument	0.411	0.578***	-0.001
	(0.509)	(0.172)	(0.014)
Anderson–Rubin p-value (no effect of the grid)	0.421	0.001	0.954
Anderson–Rubin 95% set for the effect	$[-20.7,\ 112.0]$	$[6.7,\ 89.8]$	$[-1.38,\ 1.98]$
Panel B. Largest rural parcel within 5 km (current cadastre)			
First stage: instrument	0.013	0.013	0.007
	(0.008)	(0.008)	(0.008)
Effective F (critical value 23.1)	2.38	2.40	0.79
Reduced form: instrument	0.170	-0.378**	-0.040***
	(0.391)	(0.156)	(0.013)
Anderson–Rubin p-value (no effect of the grid)	0.665	0.016	0.004
Anderson–Rubin 95% set for the effect	$(-\infty,\ \infty)$	$(-\infty,\ -4.9]$ $\cup\ [94.8,\ \infty)$	$(-\infty,\ -1.40]$ $\cup\ [3.27,\ \infty)$
Panel C. Hydro dams, without the schools within 100 km of one			
First stage: instrument	0.021*	0.021*	0.022**
	(0.011)	(0.011)	(0.010)
Effective F (critical value 23.1)	3.63	3.62	4.41
Reduced form: instrument	-0.204	0.616**	0.019
	(0.500)	(0.235)	(0.018)
Anderson–Rubin p-value (no effect of the grid)	0.684	0.010	0.308
Anderson–Rubin 95% set for the effect	$(-\infty,\ \infty)$	$(-\infty,\ -673.0]$ $\cup\ [6.5,\ \infty)$	$[-0.85,\ 20.79]$
Schools / school-years / clusters (Panel C)	421 / 11,503 / 122	421 / 11,505 / 122	421 / 11,581 / 122
Mean of the outcome (Panels A and B)	84.1	5.8	2.29
School fixed effects	Yes	Yes	Yes
Region × year fixed effects	Yes	Yes	Yes
Schools (Panels A and B)	586	586	586
Observations (school-years, Panels A and B)	16,010	16,013	16,119
Clusters (police sections, Panels A and B)	174	174	174

Table 6: Candidate instruments for the timing of grid connection: first stages, reduced forms, and Anderson–Rubin tests.

Notes: Panel A: the instrument is the standardized $\log(1 + x)$ of the km from the school to the nearest of the four hydro dams, times (year − 1993)/10. Panel B: the same with the largest rural parcel within 5 km. Panel C: Panel A without the 165 risk-set schools within 100 km of a dam. Sample: the 586 rural schools of the analysis sample without grid at the start of 1993, in all their open years from 1993 to 2021. First stage: the indicator of being on the grid on the instrument. Reduced form: the outcome on the instrument. The Anderson–Rubin 95% set collects the effects of the grid that the test does not reject, in the units of the outcome. Standard errors clustered by police section in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: own elaboration based on ANEP and UTE records, MIEM (dams), and DNC's rural cadastre.

C Additional results

This appendix collects additional results behind Section 4. Appendix Table 7 reports the main estimates in the broad sample, and Appendix Table 8 in the reduced sample. Appendix Table 9 reports the robustness of the main estimate, and Appendix Table 10 its robustness to group-specific trends. Appendix Table 11 reports the effects on enrollment, teachers, and school closure, and Appendix Table 12 those of moving from a solar panel to the grid and from a utility panel to a Lights to Learn panel. Appendix Table 13 reports the 1996 national assessment. For Section 5, Appendix Table 14 reports the split by distance to larger localities, and Appendix Figure 12 the monthly night light around five schools next to a hamlet.

	From no electricity			
	To the grid (1)	To any source (2)	To a solar panel (3)	Mean without electricity (4)
Satisfactory attendance (141 days or more)	3.95***	3.45***	1.94	83.9
	(1.23)	(1.19)	(3.55)	
	[337; 0.60]	[431; 0.34]	[93; 0.30]	
Insufficient attendance (71–140 days)	-3.93***	-3.11***	-0.23	13.7
	(1.05)	(1.08)	(2.48)	
	[337; 0.65]	[431; 0.18]	[93; 0.13]	
Intermittent dropout (70 days or fewer)	-0.01	-0.35	-1.71	2.4
	(0.46)	(0.49)	(2.40)	
	[337; 0.05]	[431; 0.06]	[93; 0.14]	
Repetition	-1.26*	-0.72	1.52	10.0
	(0.74)	(0.67)	(1.45)	
	[337; 0.01]	[431; 0.65]	[93; 0.56]	
School and year fixed effects	Yes	Yes	Yes	
Comparisons within region × year	Yes	Yes	Yes	
Controls: weather and Plan Ceibal	Yes	Yes	Yes	
Schools	576	576	576	
School-years, attendance	12,038	14,889	6,163	
School-years, repetition	12,041	14,892	6,166	

Table 7: Effects of electrifying a school on attendance and repetition.

Notes: Unit: school-year, 1992–2019; broad rural sample (1,102 schools, without the 5 percent least rural). Each cell averages the event-study effects of de Chaisemartin and D'Haultfœuille (2020, 2026) over the first two school years with electricity, in percentage points, weighted by first-year enrollment; each switching school is compared with schools of its region that also start without electricity and have not switched yet. Standard errors clustered by police section in parentheses. In brackets: switching schools; p-value of the joint test that the four placebos are zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Controls: flood days, rainfall, and cold and hot days (attendance only), and exposure to Plan Ceibal. Last column: unweighted mean of the outcome in school-years without electricity, in percent. Source: own elaboration based on ANEP and UTE records.

	From no electricity			
	To the grid	To any source	To a solar panel	Mean without electricity
	(1)	(2)	(3)	(4)
Satisfactory attendance (141 days or more)	4.36***	3.79***	2.12	83.9
	(1.32)	(1.27)	(3.61)	
	[324; 0.53]	[418; 0.42]	[93; 0.30]	
Insufficient attendance (71–140 days)	-4.41***	-3.45***	-0.32	13.8
	(1.13)	(1.16)	(2.56)	
	[324; 0.77]	[418; 0.25]	[93; 0.13]	
Intermittent dropout (70 days or fewer)	0.05	-0.34	-1.80	2.4
	(0.49)	(0.52)	(2.40)	
	[324; 0.06]	[418; 0.06]	[93; 0.15]	
Repetition	-1.44*	-0.85	1.42	10.0
	(0.79)	(0.70)	(1.45)	
	[324; 0.03]	[418; 0.73]	[93; 0.57]	
School and year fixed effects	Yes	Yes	Yes	
Comparisons within region × year	Yes	Yes	Yes	
Controls: weather and Plan Ceibal	Yes	Yes	Yes	
Schools	563	563	563	
School-years, attendance	11,691	14,542	6,111	
School-years, repetition	11,694	14,545	6,114	

Table 8: Effects of electrifying a school on attendance and repetition, reduced sample.

Notes: Unit: school-year, 1992–2019; reduced rural sample (990 schools, without the 15 percent least rural). Each cell averages the event-study effects of de Chaisemartin and D'Haultfœuille (2020, 2026) over the first two school years with electricity, in percentage points, weighted by first-year enrollment; each switching school is compared with schools of its region that also start without electricity and have not switched yet. Standard errors clustered by police section in parentheses. In brackets: switching schools; p-value of the joint test that the four placebos are zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Controls: flood days, rainfall, and cold and hot days (attendance only), and exposure to Plan Ceibal. Last column: unweighted mean of the outcome in school-years without electricity, in percent. Source: own elaboration based on ANEP and UTE records.

	Broad sample	Reduced sample
Panel A. Specification		
Main specification	3.95*** (1.23) [337; 0.60]	4.36*** (1.32) [324; 0.53]
Without controls	3.91*** (1.23) [337; 0.57]	4.33*** (1.32) [324; 0.48]
Only year effects (no region × year)	3.17*** (1.14) [342; 0.18]	3.56*** (1.22) [329; 0.08]
Comparisons within department × year	3.50*** (1.33) [326; 0.56]	3.89*** (1.42) [314; 0.58]
School-specific linear trends	4.59* (2.72) [306]	4.81 (2.94) [296]
Initial enrollment × year	4.29*** (1.23) [337; 0.79]	4.67*** (1.31) [324; 0.66]
Region × tercile of the initial outcome × year	4.00*** (1.33) [326; 0.33]	4.51*** (1.43) [313; 0.33]
One year of anticipation (reference: two years before)	4.10*** (1.44) [307; 0.46]	4.73*** (1.51) [297; 0.35]
Same switching schools at both horizons	3.88*** (1.27) [320; 0.77]	4.30*** (1.36) [307; 0.71]
Controls open in the whole window	3.94*** (1.24) [337; 0.62]	4.36*** (1.32) [324; 0.55]
Control for current enrollment (post-treatment)	3.93*** (1.24) [337; 0.62]	4.38*** (1.33) [324; 0.54]
Panel B. Weights		
Without enrollment weights	2.89** (1.27) [337; 0.91]	2.90** (1.31) [324; 0.87]
Weights capped at the 95th percentile	3.41*** (1.20) [337; 0.53]	3.65*** (1.28) [324; 0.43]
Panel C. Sample		
All schools of the analysis sample (no rurality cut)	4.39*** (1.14) [343; 0.41]	4.39*** (1.14) [343; 0.41]
Without schools with fewer than 5 students	3.95*** (1.28) [315; 0.60]	4.36*** (1.37) [302; 0.53]
Without years of temporary reclassification	3.61*** (1.23) [326; 0.71]	4.01*** (1.33) [313; 0.50]
Without years after the school stops being rural	3.92*** (1.24) [337; 0.63]	4.36*** (1.32) [324; 0.53]
Panel D. Connection dates		
Without the 12 dates that contradict a documented bound	3.62*** (1.26) [334; 0.64]	3.85*** (1.36) [321; 0.54]
Only connection dates known to the day	3.99*** (1.24) [337; 0.46]	4.40*** (1.32) [324; 0.41]
Only connection dates of high certainty	2.95* (1.64) [226; 0.66]	3.49** (1.76) [217; 0.45]
Without connection dates of low certainty	3.93*** (1.32) [326; 0.63]	4.18*** (1.43) [313; 0.59]
Without schools with no connection date	4.18*** (1.24) [337; 0.85]	4.57*** (1.32) [324; 0.79]
Without schools whose date 1996 sources make doubtful	3.81*** (1.14) [334; 0.51]	4.22*** (1.22) [321; 0.41]
Alternative date: UTE's contract date	3.59** (1.44) [231; 0.51]	3.62** (1.50) [225; 0.36]
Alternative date: next-ranked source	4.14*** (1.25) [327; 0.67]	4.58*** (1.34) [314; 0.60]
Panel E. Cohorts and departments		
Only switchers connected from 1997 on (four placebos)	2.96** (1.46) [207; 0.44]	3.58** (1.57) [201; 0.39]
Only switchers connected in 1993–2002	3.52*** (1.31) [285; 0.50]	3.83*** (1.41) [274; 0.59]
Only switchers connected from 2003 on	0.34 (3.26) [52; 0.09]	0.56 (3.45) [50; 0.11]
Leaving out one connection cohort: minimum (without the 1994 cohort)	2.38* (1.25) [298; 0.53]	2.83** (1.35) [287; 0.48]
Leaving out one connection cohort: maximum (without the 1995 cohort)	4.89*** (1.35) [313; 0.55]	5.42*** (1.46) [300; 0.44]
Leaving out one department: minimum (without Salto)	3.01** (1.20) [310; 0.69]	3.43*** (1.30) [298; 0.61]
Leaving out one department: maximum (without Tacuarembó; reduced sample, Rivera)	4.60*** (1.32) [309; 0.82]	4.91*** (1.39) [292; 0.95]
Panel F. Window		
Window extended to 2021, without 2020	3.95*** (1.23) [337; 0.60]	4.36*** (1.32) [324; 0.53]
Window extended to 2021	3.95*** (1.23) [337; 0.60]	4.36*** (1.32) [324; 0.53]
Panel G. Inference		
Standard errors clustered by school	3.95*** (1.21) [337; 0.63]	4.36*** (1.29) [324; 0.57]
Pairs cluster bootstrap by police section	3.95*** (1.49) [337; 0.60] 95% interval: 0.98 to 6.80	4.36*** (1.65) [324; 0.53] 95% interval: 1.19 to 7.64
Pairs cluster bootstrap by department	3.95*** (1.52) [337; 0.60] 95% interval: 0.84 to 6.54	4.36*** (1.46) [324; 0.53] 95% interval: 0.85 to 6.70
Panel H. Other estimators		
Imputation (Borusyak, Jaravel, and Spiess)	3.76** (1.68) [343]	4.22*** (1.46) [330]
Stacked difference-in-differences	3.31** (1.32) [349]	3.71*** (1.39) [336]
Sun and Abraham (unweighted)	2.10 (not interpretable)	2.12 (not interpretable)
Two-way fixed effects (reference only)	2.14* (1.26) [349]	2.58* (1.33) [336]
Panel I. Sensitivity to violations of parallel trends		
Rambachan and Roth: breakdown value of $\bar{M}$	$\bar{M} = 0.41$	$\bar{M} = 0.38$

Table 9: Robustness of the effect of a grid connection on satisfactory attendance.

Notes: Outcome: share of students with satisfactory attendance (141 days or more). Design: each school that switches from no electricity to the grid is compared with schools of its region that start without electricity and have not switched yet, as in Table 7. Each row changes one element of the main specification. Cells: average effect over the first two school years with electricity, in percentage points; standard errors clustered by police section in parentheses; in brackets, switching schools and the p-value of the joint placebo test (linear trends and the other estimators have none). Bootstrap: 999 draws by police section and 199 by department, with the 95% percentile interval. The Sun–Abraham standard error is not interpretable. $\bar{M}$: largest deviation from parallel trends, relative to the pre-treatment ones, that keeps the effect significant. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: own elaboration based on ANEP and UTE records.

	Broad sample		Reduced sample	
	To the grid	To any source	To the grid	To any source
Main specification (region × year)	3.95*** (1.23)	3.45*** (1.19)	4.36*** (1.32)	3.79*** (1.27)
Panel A. Trends by initial characteristics and isolation				
1. Region × isolated × year	4.11*** (1.33)	3.49*** (1.34)	4.42*** (1.41)	3.67*** (1.42)
2a. Trend: distance to a locality of 5,000 inhabitants or more	3.93*** (1.23)	3.44*** (1.19)	4.30*** (1.32)	3.75*** (1.27)
2b. Trend: population within 5 km, 1990	4.02*** (1.23)	3.50*** (1.19)	4.36*** (1.32)	3.79*** (1.27)
2c. Trend: 1992 enrollment	4.28*** (1.23)	3.69*** (1.19)	4.65*** (1.31)	3.95*** (1.27)
2d. Trends 2a–2c together	4.27*** (1.24)	3.68*** (1.20)	4.57*** (1.31)	3.90*** (1.27)
2t. Region × tercile of distance × year	4.55*** (1.33)	3.91*** (1.32)	4.76*** (1.41)	4.07*** (1.40)
2t. Region × tercile of population × year	4.08*** (1.14)	3.21*** (1.16)	4.63*** (1.27)	3.63*** (1.28)
2t. Region × tercile of enrollment × year	4.43*** (1.61)	3.80*** (1.42)	4.92*** (1.70)	4.15*** (1.48)
5. Variants 1 and 2d together	4.35*** (1.34)	3.59*** (1.33)	4.65*** (1.41)	3.79*** (1.41)
Panel B. Trends by the 1996 context of the census zone (schools with a census zone)				
Main specification, schools of 3a–3c	3.51** (1.52)	2.91* (1.59)	3.72** (1.61)	3.05* (1.66)
3a. Trend: adult education, 1996	3.51** (1.52)	2.91* (1.57)	3.74** (1.61)	3.06* (1.64)
3b. Trend: housing unmet basic needs, 1996	3.51** (1.52)	2.91* (1.58)	3.73** (1.61)	3.06* (1.65)
3c. Trends 3a and 3b together	3.51** (1.52)	2.91* (1.57)	3.74** (1.61)	3.06* (1.64)
Main specification, schools of 3d	2.32 (1.93)	1.77 (2.06)	2.53 (2.00)	1.91 (2.11)
3d. As 3c, without switchers by 1996	2.33 (1.92)	1.77 (2.03)	2.56 (2.00)	1.92 (2.08)
5b. All together (1, 2d, and 3c)	3.52* (1.82)	2.71 (1.78)	3.66* (1.92)	2.77 (1.86)
Panel C. A covariate measured mostly after connection				
4. Trend: 2005 school unmet basic needs index	3.53*** (1.26)	3.27*** (1.20)	3.96*** (1.36)	3.63*** (1.29)
School and year fixed effects	Yes	Yes	Yes	Yes
Comparisons within region × year (finer in 1, 2t, 5, and 5b)	Yes	Yes	Yes	Yes
Weather and Plan Ceibal controls	Yes	Yes	Yes	Yes
Switching schools, main specification	337	431	324	418
Distinct controls, main specification	510	510	501	501
Schools, main specification	576	576	563	563

Table 10: Robustness of the attendance effect to group-specific trends.

Notes: Average effect of electrifying a school on the share of students with satisfactory attendance (141 days or more) over its first two school years with electricity, in percentage points, estimated as in Table 7; standard errors clustered by police section in parentheses. Columns: from no electricity to the grid, or to any source (grid or solar panel). Each numbered row lets schools follow different trends: linear in the standardized log of the characteristic, or comparisons within terciles (2t). Isolated: no locality within 1 km in the 2011 census; adult education: mean years of schooling, ages 25–64. Panel B uses the schools with a 1996 census zone (488 broad, 480 reduced; 381 and 378 in 3d); the italic rows show that its lower estimates come from the sample, not the trends. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: own elaboration based on ANEP and UTE records, INE's 1996 and 2011 censuses, and GHSL.

	From no electricity			
	To the grid (1)	To any source (2)	To a solar panel (3)	Mean without electricity (4)
Enrollment, grades 1–6 (log points × 100)	2.56	1.87	1.21	2.39
	(2.32)	(2.91)	(6.14)	
	[357; 0.33]	[449; 0.39]	[91; 0.80]	
Teachers, grades 1–6	-0.00	-0.02	-0.07	1.22
	(0.02)	(0.03)	(0.07)	
	[323; 0.37]	[415; 0.59]	[91; 0.69]	
Students per teacher	0.30	0.33	0.45	9.82
	(0.25)	(0.25)	(0.48)	
	[323; 0.29]	[414; 0.29]	[90; 0.72]	
School open (percentage points)	0.63	1.54	3.89	98.2
	(0.48)	(1.16)	(3.27)	
	[360; 0.04]	[452; 0.17]	[91; 0.35]	
Teacher experience (connections from 2003 on)				
Share of teachers with less than 5 years' experience	-0.25†	1.10	2.95	16.1
	(6.57)	(5.21)	(6.00)	
	[46; 0.50]	[116; 0.22]	[70; 0.16]	
Share of teachers with less than 3 years in the school	-6.02	5.18	10.68	69.4
	(9.07)	(7.41)	(8.55)	
	[49; 0.42]	[121; 0.65]	[72; 0.43]	
School and year fixed effects	Yes	Yes	Yes	
Comparisons within region × year	Yes	Yes	Yes	
Control: Plan Ceibal	Yes	Yes	Yes	
Schools; school-years				
Enrollment	590; 13,366	590; 16,303	590; 6,680	
Teachers	590; 10,242	590; 12,248	589; 6,167	
Students per teacher	590; 10,224	590; 12,215	589; 6,152	
School open	590; 13,601	590; 16,900	590; 6,949	
Experience	523; 4,254	582; 5,736	274; 1,822	
Years in the school	524; 4,279	582; 5,775	275; 1,837	

Table 11: Effects of electrifying a school on enrollment, teachers, and school closure.

Notes: Broad rural sample (1,102 schools, without the 5 percent least rural). Cells average the event-study effects of de Chaisemartin and D'Haultfœuille (2020, 2026) over the first five school years with electricity, counted from the year after the connection, since enrollment and staffing are set when the school year starts; unweighted. Teachers and students per teacher in levels; shares in percentage points. School open is descriptive: schools about to close may never get connected. Windows: enrollment and school open 1992–2021; teachers 1992–2015, without 2002–2003; experience 2002–2013. Standard errors clustered by police section in parentheses; brackets: switching schools; joint placebo p-value. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † A horizon has under 30 switching schools or 10 distinct controls. Column (4): unweighted mean without electricity (enrollment: log of 1 + students). Source: own elaboration based on ANEP and UTE records.

	Effect		Mean before the change	
	From a solar panel to the grid (1)	From a UTE panel to a Lights to Learn panel (2)	From a solar panel to the grid (3)	From a UTE panel to a Lights to Learn panel (4)
Satisfactory attendance (141 days or more)	1.16	18.94†	88.2	88.3
	(2.20)	(11.68)		
	[163; 0.00]	[48; 0.36]		
Counting panel years of unknown grid status‡		10.18**		88.7
		(4.99)		
		[73; 0.70]		
		{0.464; 0.052}		
Insufficient attendance (71–140 days)	0.45	-16.32†	9.5	9.2
	(2.00)	(12.53)		
	[163; 0.00]	[48; 0.45]		
Intermittent dropout (70 days or fewer)	-1.60	-2.62†	2.3	2.5
	(1.30)	(2.27)		
	[163; 0.03]	[48; 0.58]		
Repetition	-0.58	-0.30†	5.7	6.1
	(1.31)	(2.83)		
	[163; 0.28]	[48; 0.79]		
Enrollment and teachers: first five school years, unweighted				
Enrollment, grades 1–6 (log points × 100)	-3.54	-75.21†	1.98	2.08
	(4.62)	(72.94)		
	[165; 0.00]	[51; 0.71]		
Teachers, grades 1–6	-0.02	0.00†	1.11	1.12
	(0.01)	(0.00)		
	[114; 0.27]	[47; 0.63]		
Students per teacher	0.04	-1.23**†	7.31	7.48
	(0.32)	(0.52)		
	[113; 0.35]	[47; 0.53]		
School and year fixed effects	Yes	Yes		
Comparisons within region × year	Yes	Yes		
Controls: weather (attendance) and Plan Ceibal	Yes	Yes		
Schools; school-years				
Attendance and repetition	255; 3,766	254; 2,345		
Counting panel years of unknown grid status		263; 2,660		
Enrollment	251; 3,884	250; 2,325		
Teachers	251; 2,650	246; 1,924		

Table 12: Effects of moving from a solar panel to the grid, and from a UTE panel to a Lights to Learn panel.

Notes: Broad rural sample; attendance and repetition 1992–2019, enrollment 1992–2021, teachers 1992–2015 without 2002–2003. Column (1) starts from UTE's or Lights to Learn's panels; column (2) stops at the grid. Comparisons: not-yet-switched schools, same region and source. Cells average event-study effects of de Chaisemartin and D'Haultfœuille (2020, 2026) over the first two school years after the change (attendance and repetition: percentage points, weighted by first-year enrollment) or the first five, from the year after (other rows, unweighted). Police-section clustered standard errors in parentheses; brackets: switching schools; joint placebo p-value. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. † A horizon has under 30 switching schools or 10 distinct controls, or is not estimable. ‡ Also counting panel years of unknown grid status, chosen after seeing its result; braces: pairs bootstrap and permutation p-values. Columns (3)–(4): unweighted pre-change means (enrollment: log of 1 + students). Source: own elaboration based on ANEP and UTE records.

	(1)	(2)	(3)	(4)
Panel A. Reading				
Connected 1992–1996	0.167	0.194	0.166	0.043
	(0.183)	(0.171)	(0.163)	(0.181)
	[0.376; 0.384]	[0.277; 0.299]	[0.345; 0.353]	[0.828; 0.825]
Connected 1995–1996	0.410**	0.407*	0.350	0.458
	(0.173)	(0.201)	(0.197)	(0.325)
	[0.038; 0.042]	[0.076; 0.093]	[0.112; 0.134]	[0.306; 0.256]
Panel B. Math				
Connected 1992–1996	0.050	0.116	0.066	0.269*
	(0.192)	(0.184)	(0.177)	(0.152)
	[0.803; 0.799]	[0.566; 0.552]	[0.736; 0.731]	[0.090; 0.115]
Connected 1995–1996	0.066	0.173	0.127	0.394
	(0.219)	(0.215)	(0.206)	(0.317)
	[0.774; 0.779]	[0.463; 0.485]	[0.589; 0.607]	[0.371; 0.345]
Sixth-grade outcomes 1992–1994	No	Yes	Yes	Yes
Household characteristics	No	No	Yes	Yes
Region fixed effects	No	No	No	Yes

Table 13: Sixth-grade scores in the 1996 national assessment: rural schools connected to the grid and schools without electricity.

Notes: Student-level OLS. Outcome: the score in the 1996 national assessment of sixth graders, in standard deviations of the students tested. Coefficient: students of rural schools connected to the grid in 1992–1996 (or 1995–1996) minus those of rural schools with no electricity in 1996. Controls: the school's sixth-grade enrollment, repetition, and attendance and its grades 1–6 enrollment in 1992–1994; the student's sex, mother's education, people per room, siblings, and child work, from the assessment questionnaires. Standard errors clustered by school in parentheses; brackets: wild cluster bootstrap p-value and permutation p-value of the t statistic. Stars from the wild bootstrap: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Students in schools (connected schools): reading, 295 in 43 (27) and 171 in 25 (9); math, 303 in 43 (27) and 179 in 25 (9). Source: own elaboration based on ANEP and UTE records and ANEP's 1996 national assessment.

	Farther (1)	Closer (2)	Farther minus closer (3)	p-value, analytic (4)	p-value, bootstrap (5)	p-value, Holm (6)
Panel A. Broad rural sample (without the 5 percent least rural)						
Distance to the nearest locality of 5,000+ inhabitants, 1996 census (split at 23.4 km)	6.60	1.06	5.53**	0.036	0.109	0.360
	(1.86)	(1.85)	(2.64)			
	[183; 0.94]	[148; 0.43]				
Distance to the nearest locality of 1,000+ inhabitants, 2011 census (split at 13.3 km)	7.36	1.50	5.86**	0.024	0.067	0.262
	(1.94)	(1.60)	(2.59)			
	[186; 0.60]	[144; 0.96]				
Panel B. Reduced rural sample (without the 15 percent least rural)						
Distance to the nearest locality of 5,000+ inhabitants, 1996 census (split at 23.4 km)	6.60	0.23	6.36**	0.022	0.069	0.222
	(1.86)	(2.05)	(2.78)			
	[183; 0.94]	[136; 0.43]				
Distance to the nearest locality of 1,000+ inhabitants, 2011 census (split at 13.3 km)	7.80	0.92	6.88**	0.014	0.037	0.154
	(2.02)	(1.78)	(2.80)			
	[184; 0.56]	[134; 1.00]				
School and year fixed effects	Yes	Yes				
Comparisons within region × year	Yes	Yes				
Controls: weather and Plan Ceibal	Yes	Yes				

Table 14: Effect of a grid connection on satisfactory attendance, by distance to the nearest locality.

Notes: Exploratory analysis, registered before estimation. Each row splits the schools at the median distance among the 1,159 schools of the analysis sample (farther: above the median) and compares switching schools only with controls on the same side. Cells average the event-study effects of de Chaisemartin and D'Haultfœuille (2020, 2026) on the share of students with satisfactory attendance over the first two school years with electricity, in percentage points, weighted by first-year enrollment, as in Appendix Table 7. Standard errors clustered by police section in parentheses; the difference sums both groups' influence functions within police sections. Brackets: switching schools; joint placebo p-value. Stars use the analytic p-value of the difference: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Bootstrap: pairs bootstrap by police section, 999 draws. Holm: adjusted for the 11 heterogeneity splits we tested. Source: own elaboration based on ANEP and UTE records and INE's 1996 and 2011 censuses.

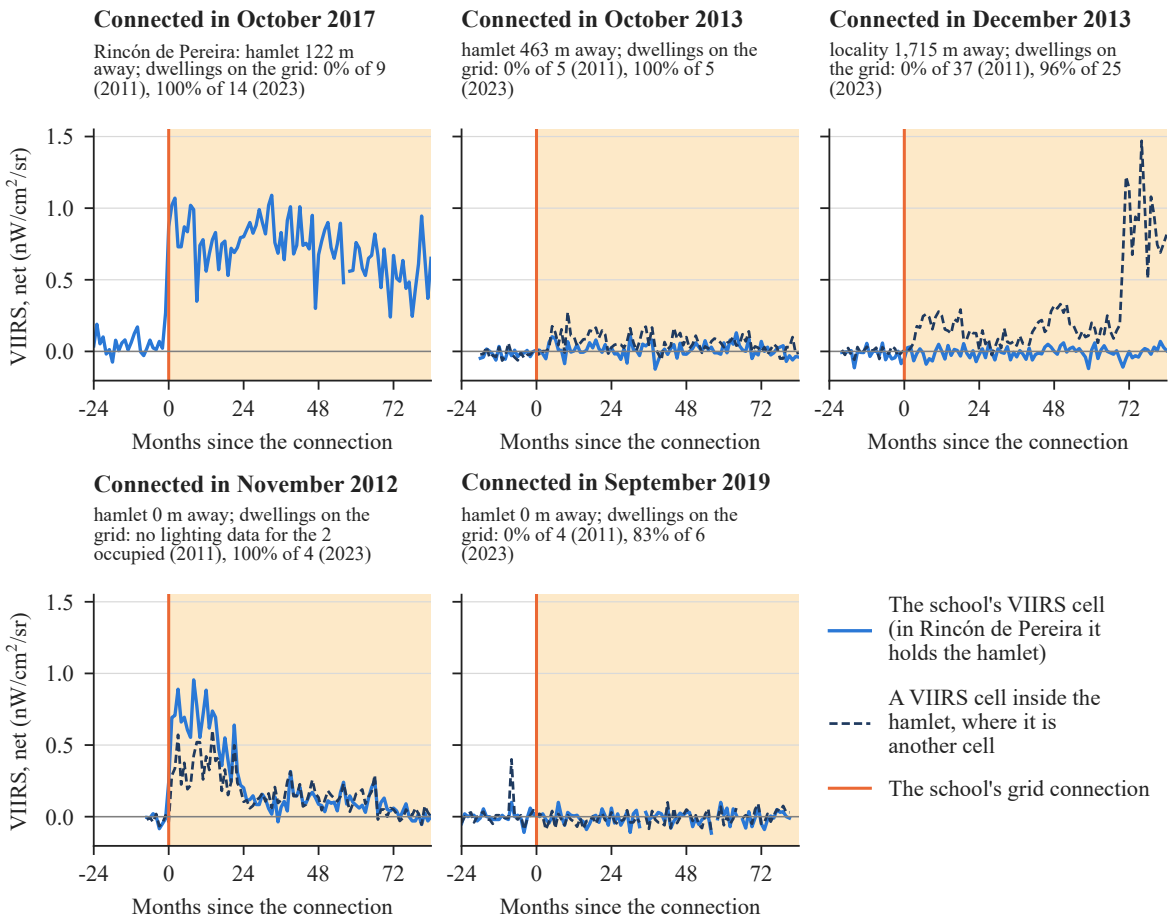


Figure 12: Night light at five rural schools next to a small census locality, around the school's grid connection.

Notes: Monthly radiance of the satellite cell over the school and, where the locality lies in another cell, over the locality, net of the median of 66 isolated schools with only a solar panel, in nW/cm^2/sr; month 0 is the school's grid connection. Rincón de Pereira is the reference case; its connection date has low certainty, and its cell lit up some weeks before the recorded date. The other four come from a rule written before looking at the light (within 1 km of a locality of 5 to 20 occupied dwellings in 2011, connected in 2012–2021 with a high-certainty date), relaxed step by step; a last step, added after a first look at the light, admits the two schools that stand inside their locality. Dwellings on the grid: share of the locality's occupied dwellings. The satellite passes after midnight and does not see the school in use. Source: own elaboration based on ANEP and UTE records, VIIRS monthly composites (Elvidge et al., 2013), and INE's 2011 and 2023 censuses.